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(54) WASHING MACHINE AND METHOD FOR CONTROLLING THE SAME

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(57) **ABSTRACT**

A washing machine may include: a tub; a detergent container configured to accommodate detergent to be supplied to the tub; a plurality of electrodes in the detergent container, a user interface configured to provide information to a user; and a controller configured to, with detergent accommodated in the detergent container: apply a current having a defined frequency to electrodes of the plurality of electrodes, detect a voltage between electrodes of the plurality of electrodes, determine an electrical conductivity of the detergent based on the detected voltage, and provide, based on the determined electrical conductivity, information about a state of the detergent through the user interface and/or adjust an amount of the detergent to be supplied to the tub.

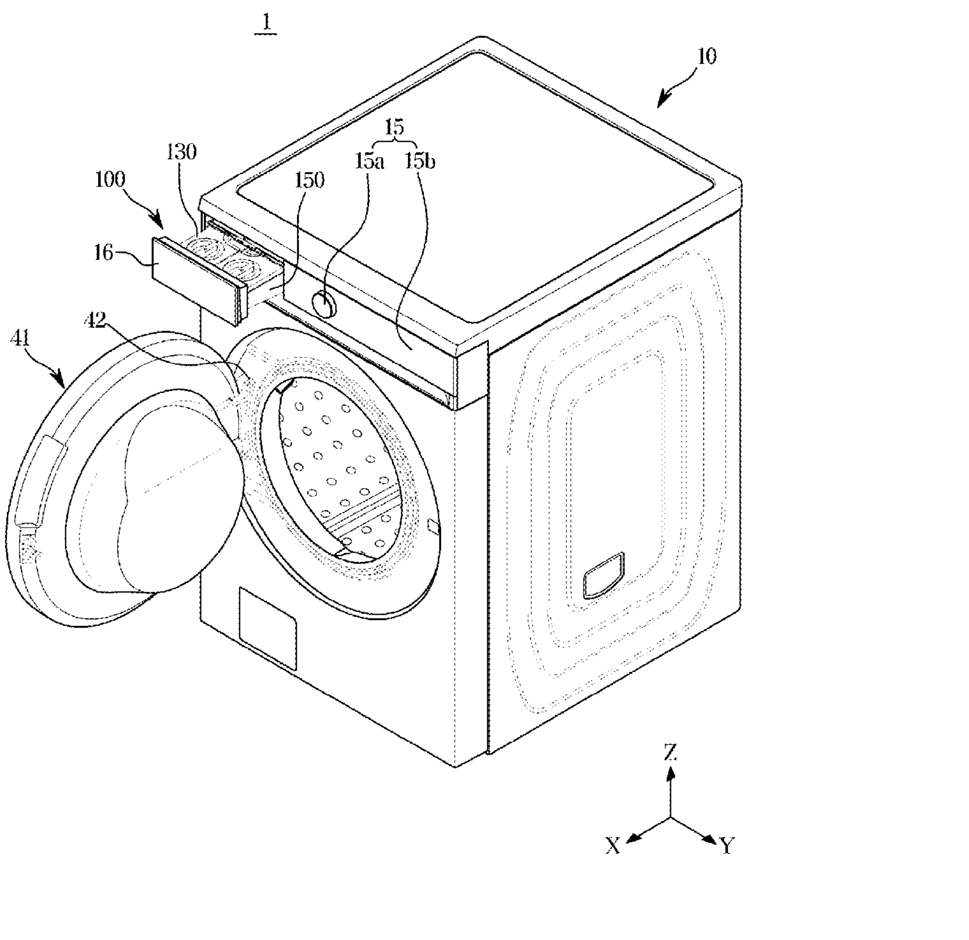


FIG. 1

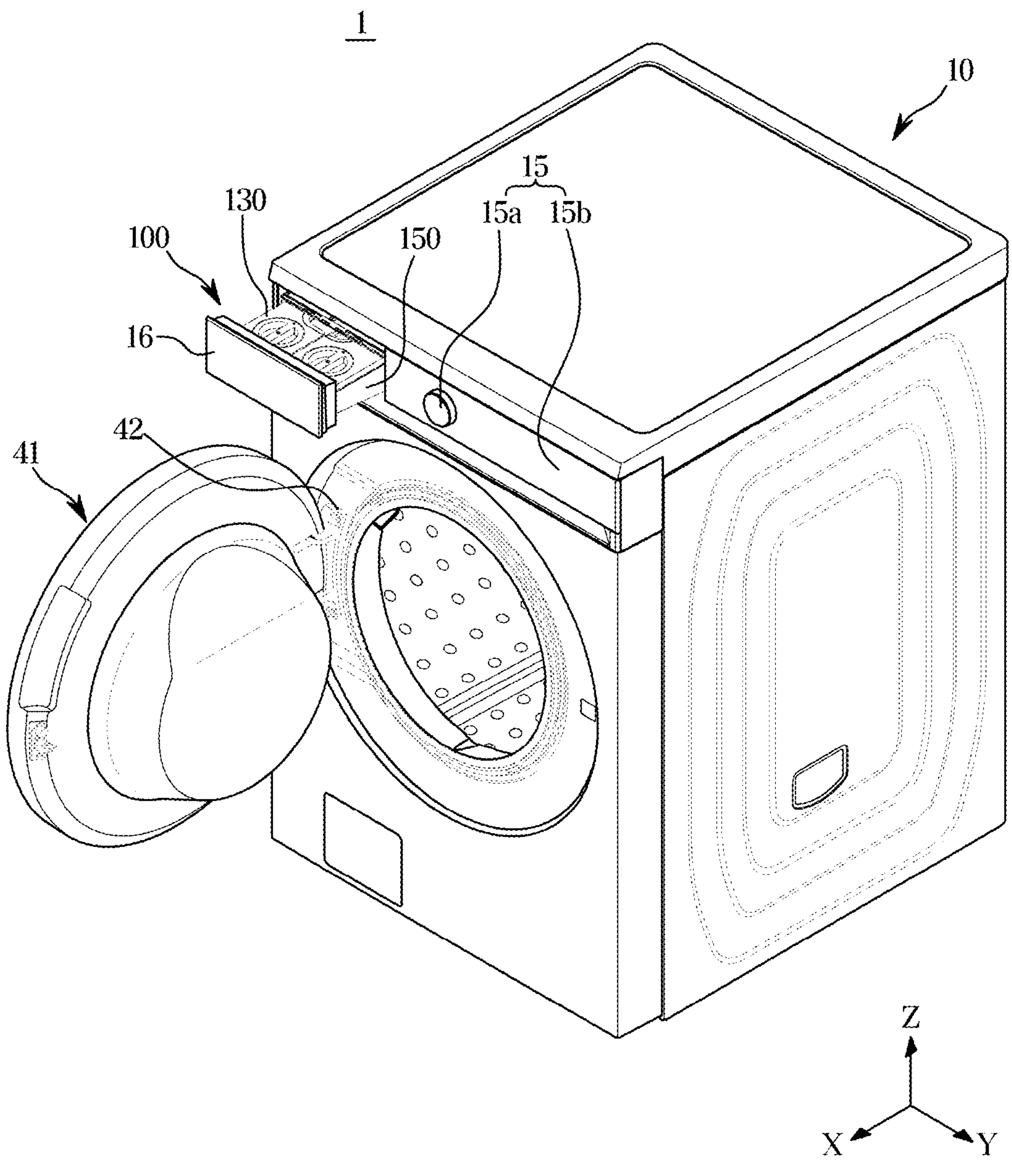


FIG. 2

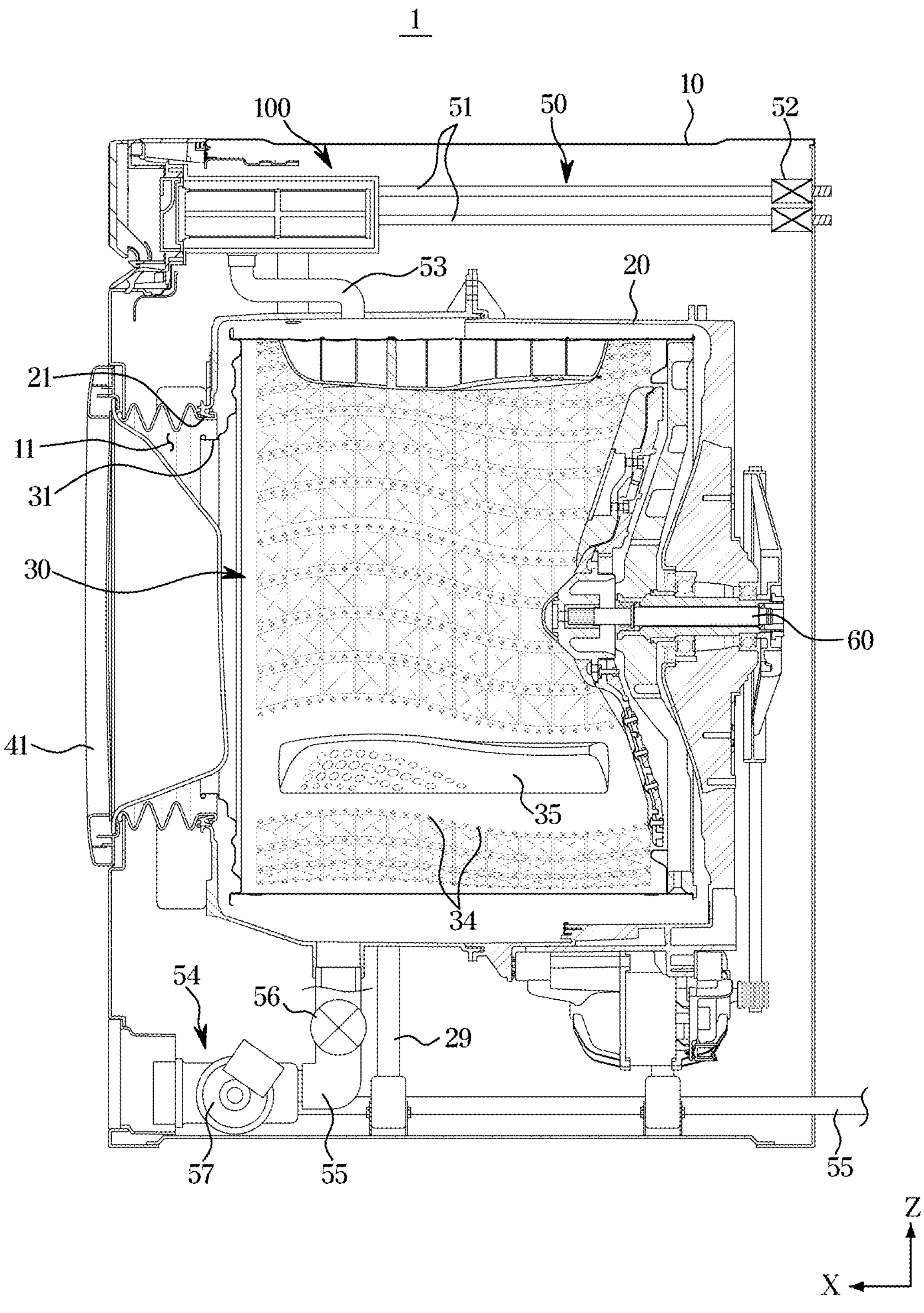
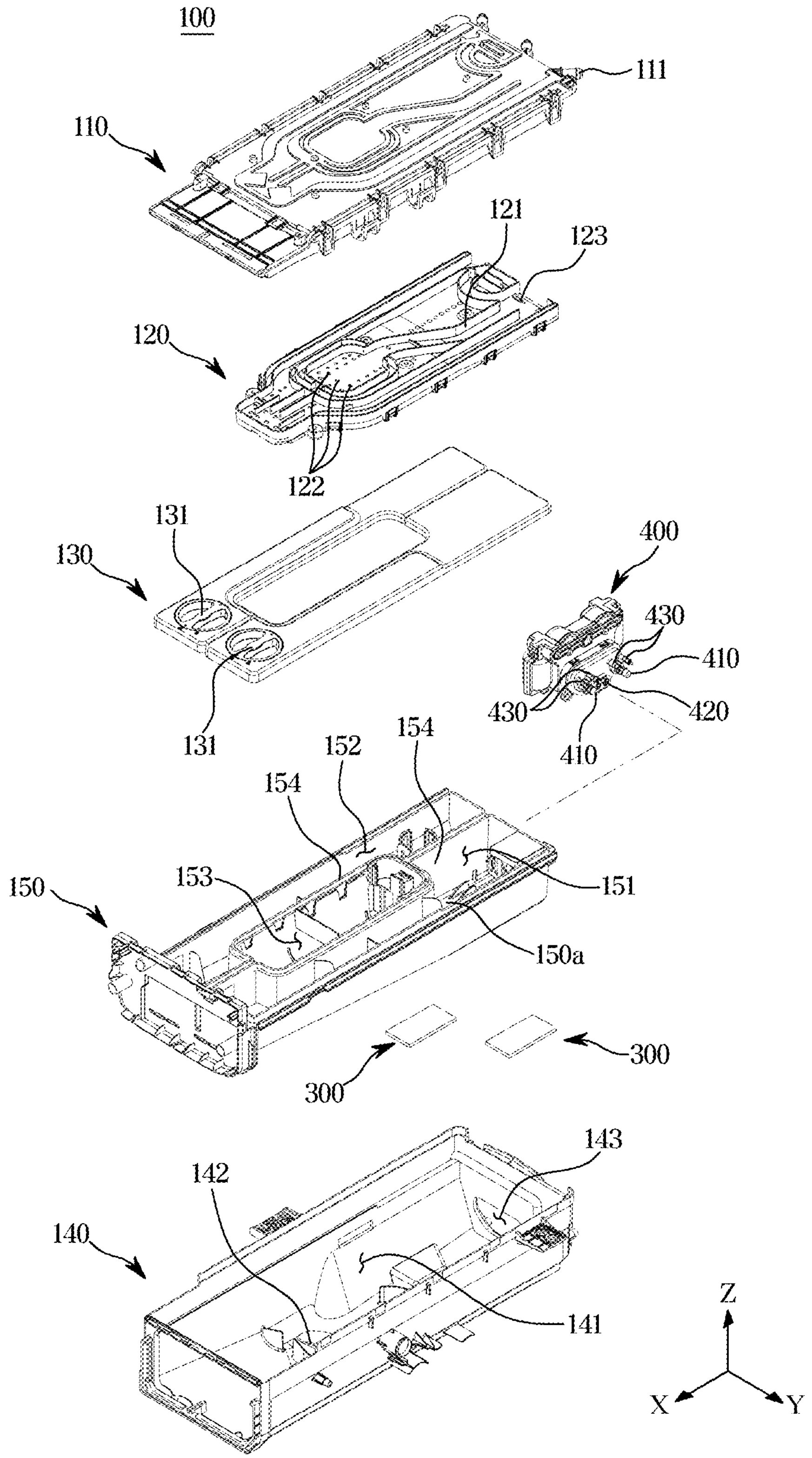
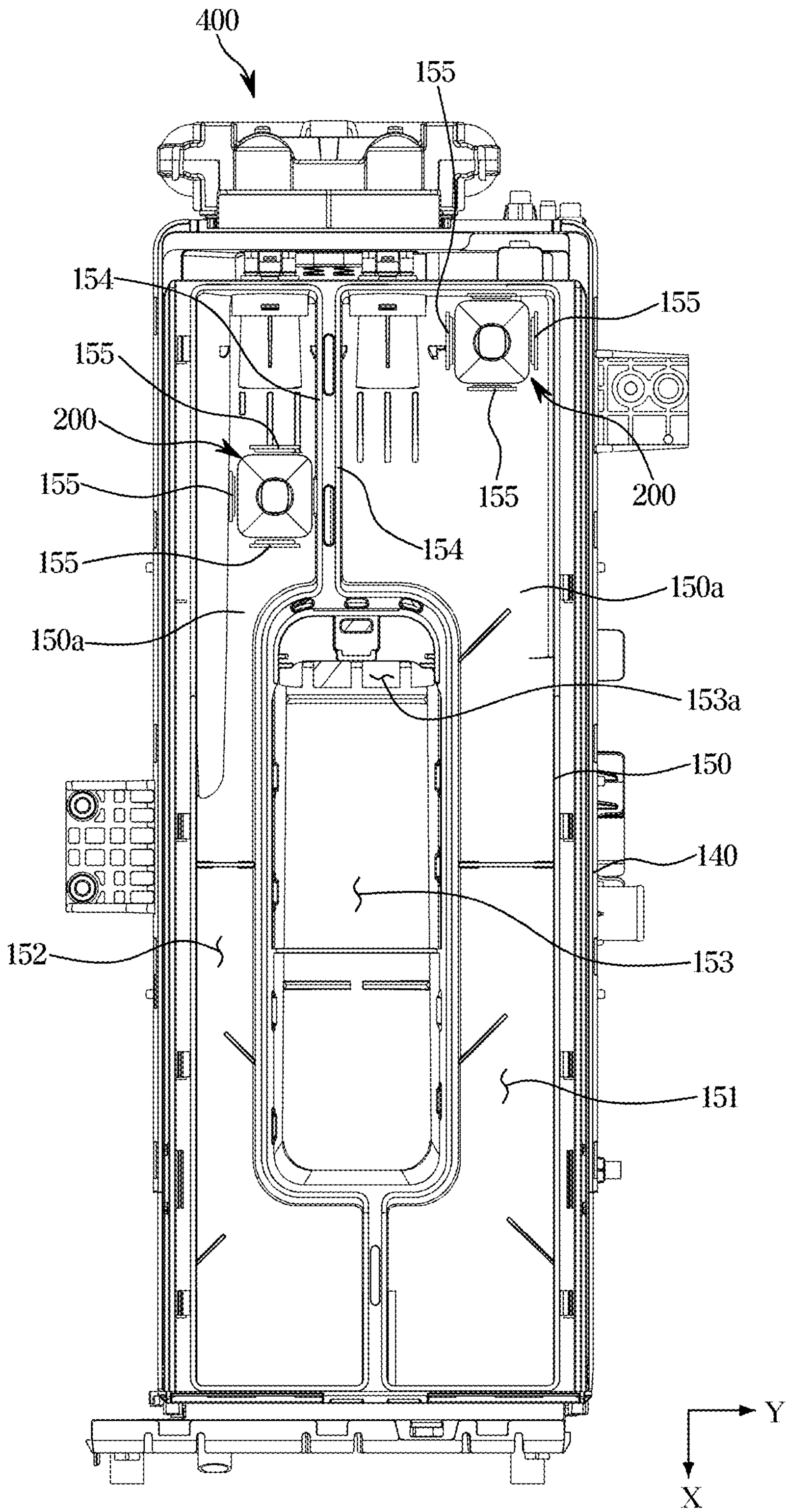


FIG. 3



100
111
110
121
123
120
122
131
130
131
400
430
410
430
420
410
152
154
154
153
151
150
150a
300
300
143
142
140
141
Z
X
Y

FIG. 4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155
154
155
155
200
155
154
200
155
155
150a
150a
153a
150
140
152
153
151
Y
X

FIG. 5

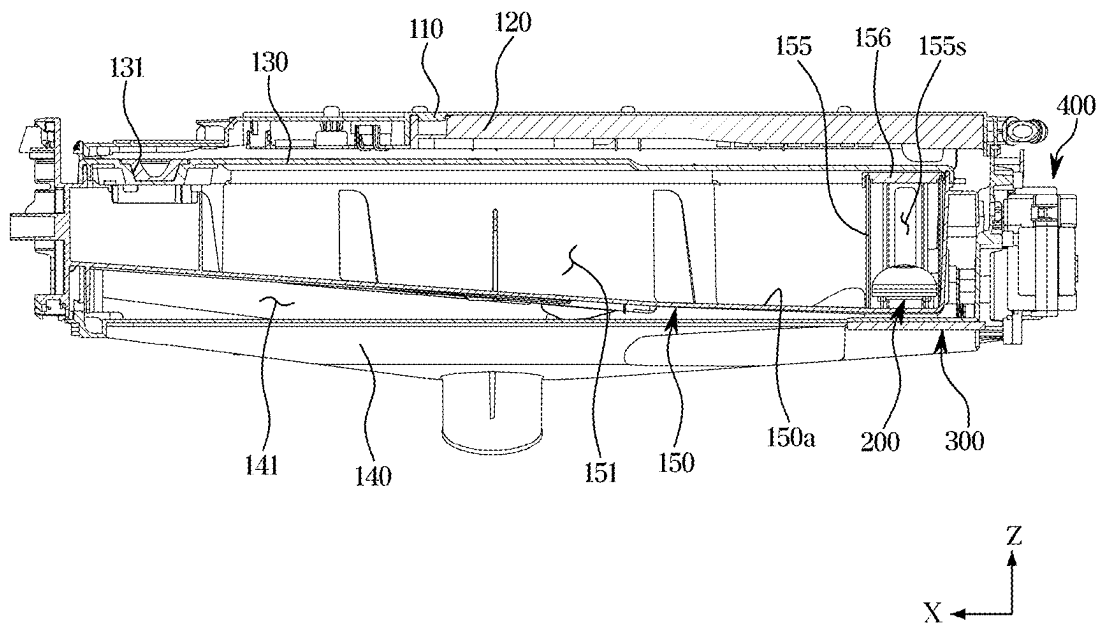

FIG. 6

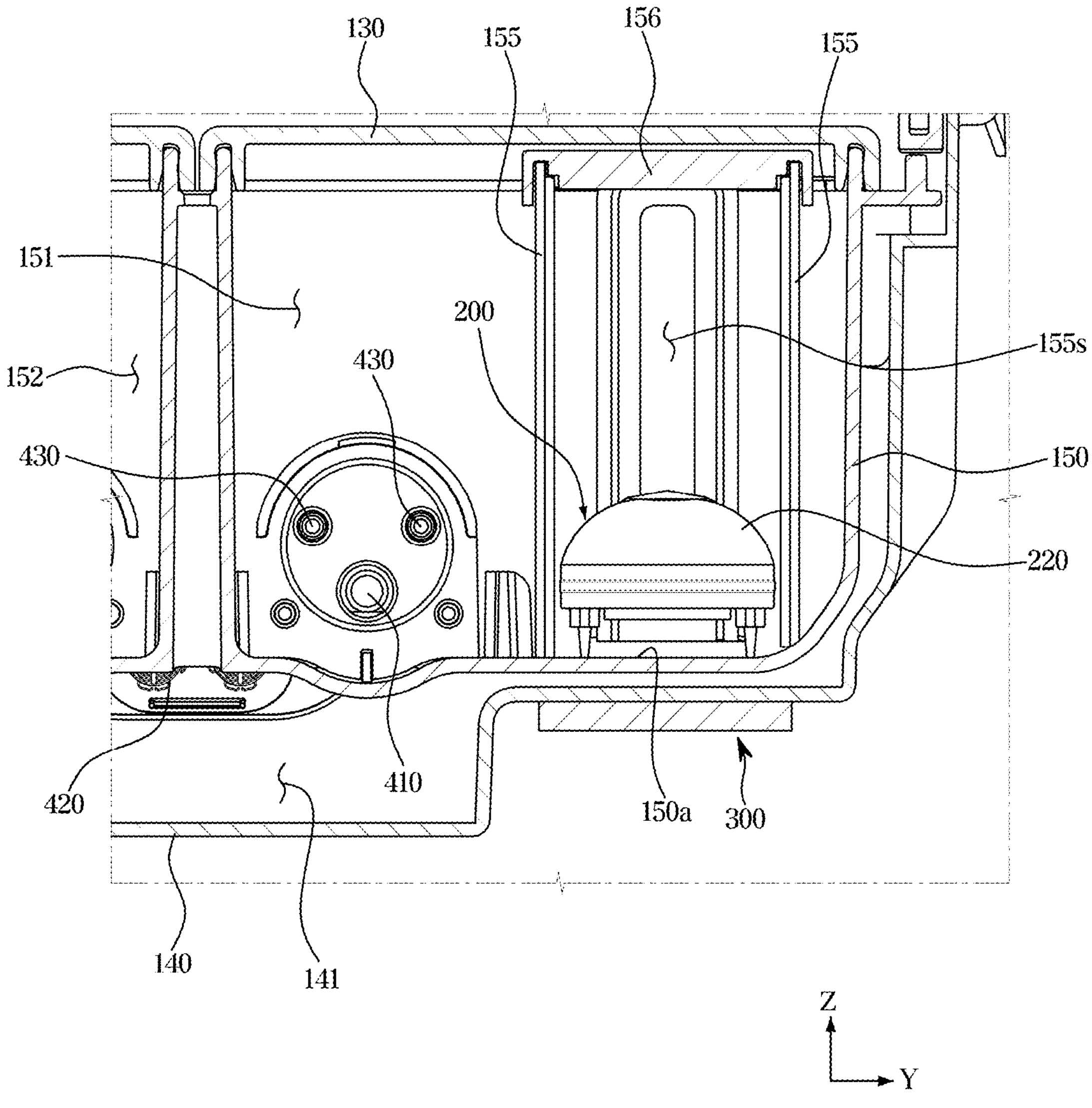
130
155
156
155
151
200
155s
152
430
430
150
220
420
410
150a
300
140
141
Z
Y

FIG. 7

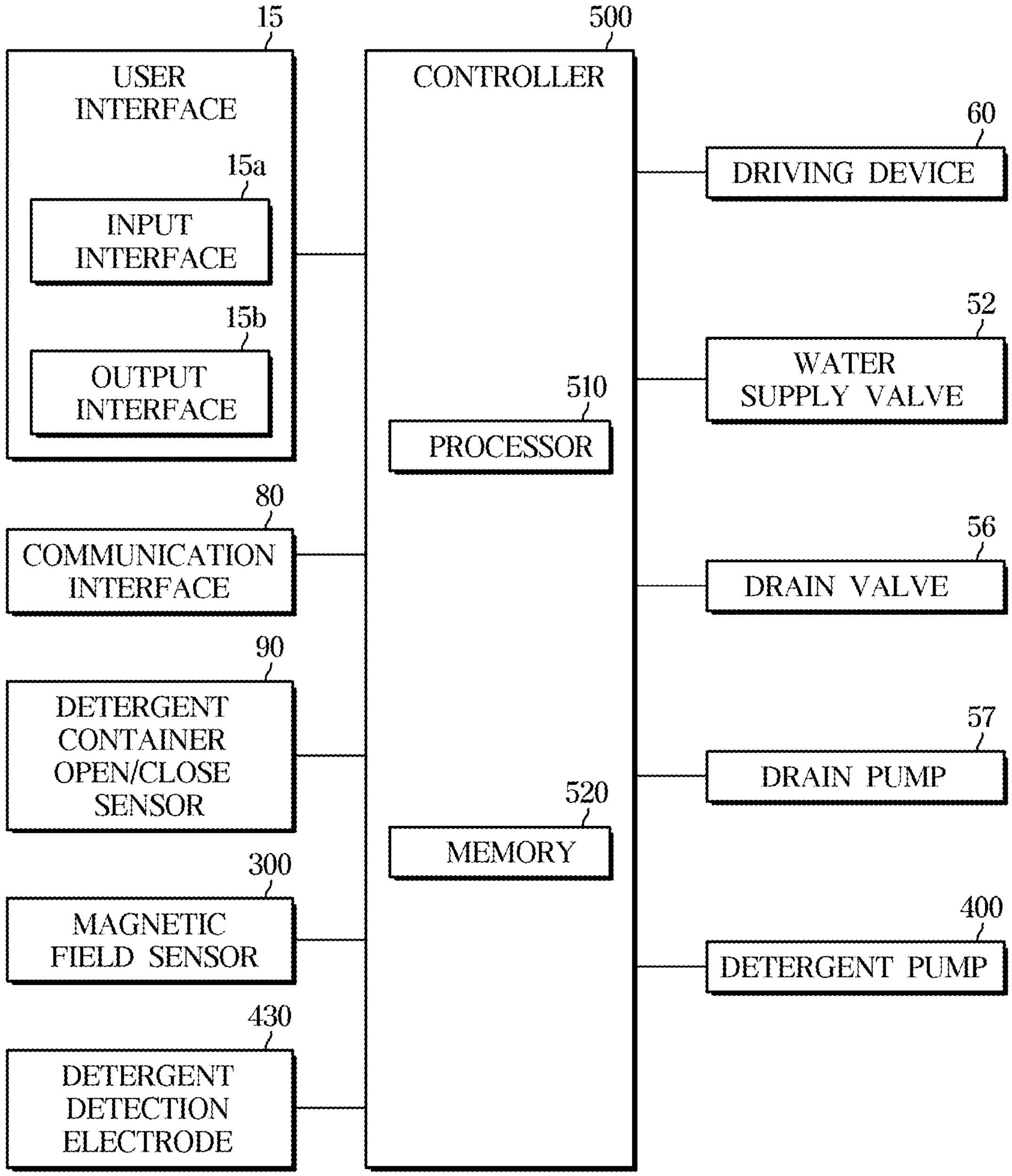
15
USER INTERFACE
15a
INPUT INTERFACE
15b
OUTPUT INTERFACE
80
COMMUNICATION INTERFACE
90
DETERGENT CONTAINER OPEN/CLOSE SENSOR
300
MAGNETIC FIELD SENSOR
430
DETERGENT DETECTION ELECTRODE
500
CONTROLLER
510
PROCESSOR
520
MEMORY
60
DRIVING DEVICE
52
WATER SUPPLY VALVE
56
DRAIN VALVE
57
DRAIN PUMP
400
DETERGENT PUMP

FIG. 8

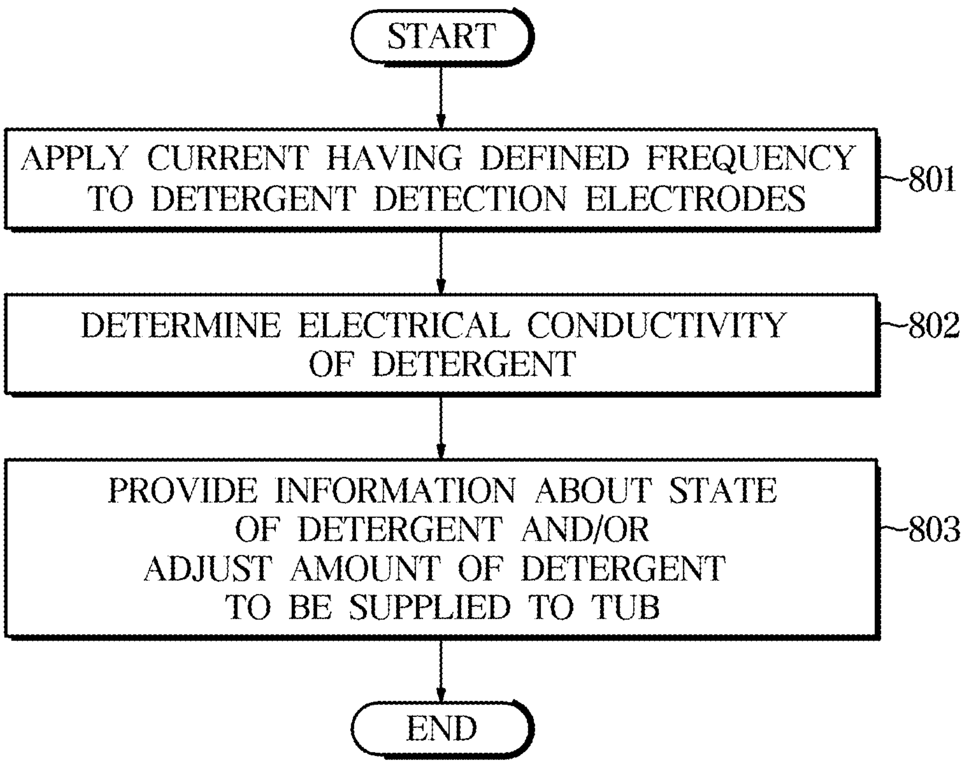
START
APPLY CURRENT HAVING DEFINED FREQUENCY TO DETERGENT DETECTION ELECTRODES
801
DETERMINE ELECTRICAL CONDUCTIVITY OF DETERGENT
802
PROVIDE INFORMATION ABOUT STATE OF DETERGENT AND/OR ADJUST AMOUNT OF DETERGENT TO BE SUPPLIED TO TUB
803
END

FIG. 9

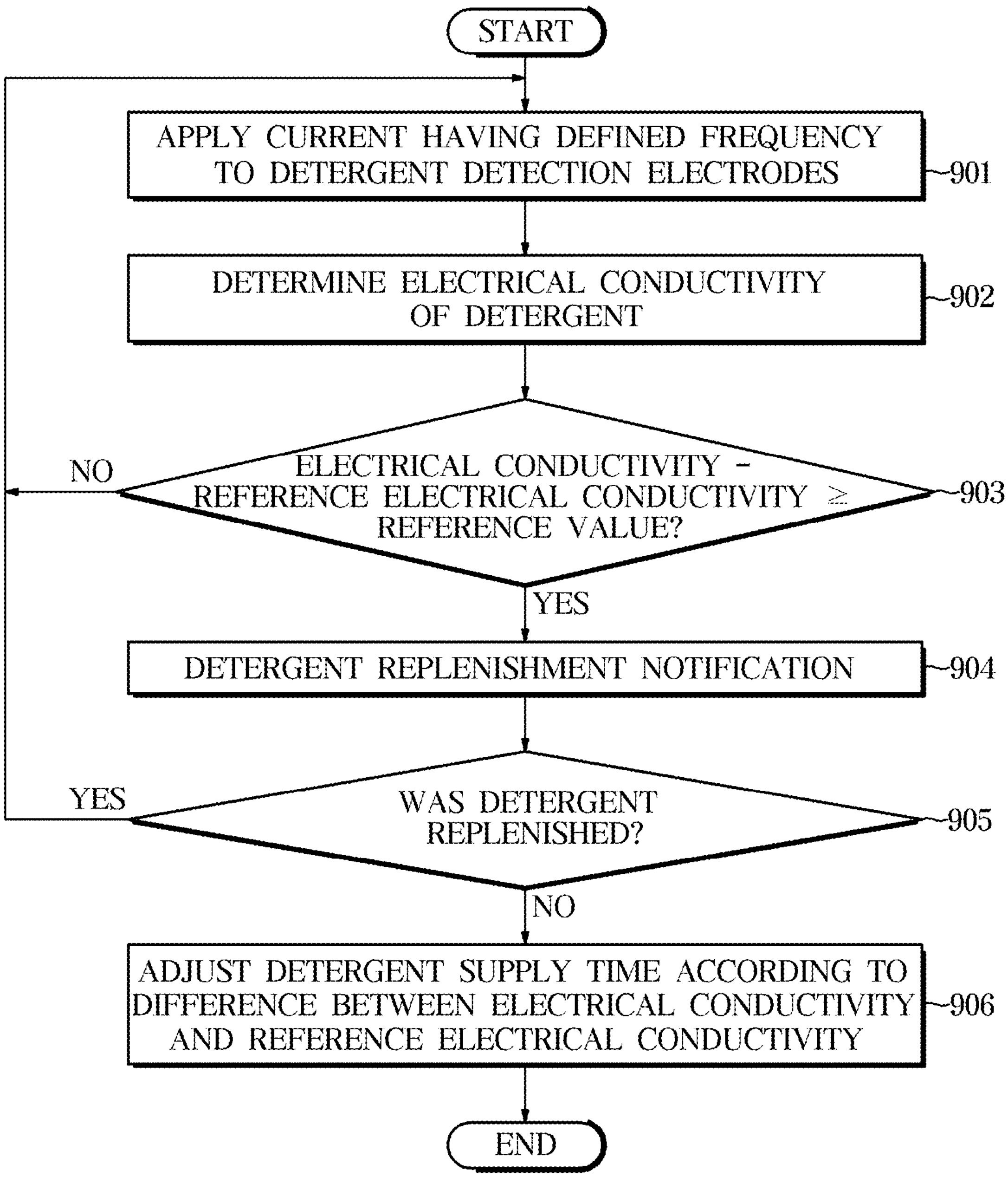
START
APPLY CURRENT HAVING DEFINED FREQUENCY TO DETERGENT DETECTION ELECTRODES
901
DETERMINE ELECTRICAL CONDUCTIVITY OF DETERGENT
902
NO
ELECTRICAL CONDUCTIVITY − REFERENCE ELECTRICAL CONDUCTIVITY ≥ REFERENCE VALUE?
903
YES
DETERGENT REPLENISHMENT NOTIFICATION
904
YES
WAS DETERGENT REPLENISHED?
905
NO
ADJUST DETERGENT SUPPLY TIME ACCORDING TO DIFFERENCE BETWEEN ELECTRICAL CONDUCTIVITY AND REFERENCE ELECTRICAL CONDUCTIVITY
906
END

FIG. 10

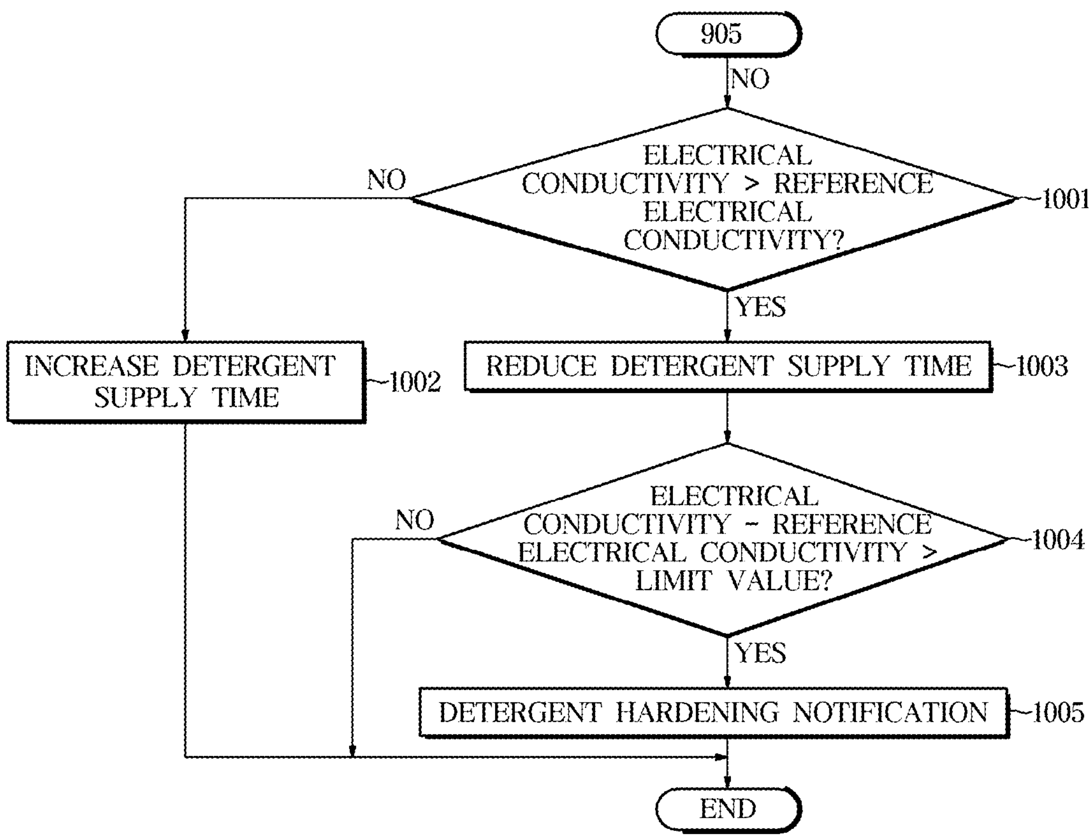
905
NO
ELECTRICAL CONDUCTIVITY > REFERENCE ELECTRICAL CONDUCTIVITY?
1001
NO
YES
INCREASE DETERGENT SUPPLY TIME
1002
REDUCE DETERGENT SUPPLY TIME
1003
ELECTRICAL CONDUCTIVITY - REFERENCE ELECTRICAL CONDUCTIVITY > LIMIT VALUE?
1004
NO
YES
DETERGENT HARDENING NOTIFICATION
1005
END

FIG. 11

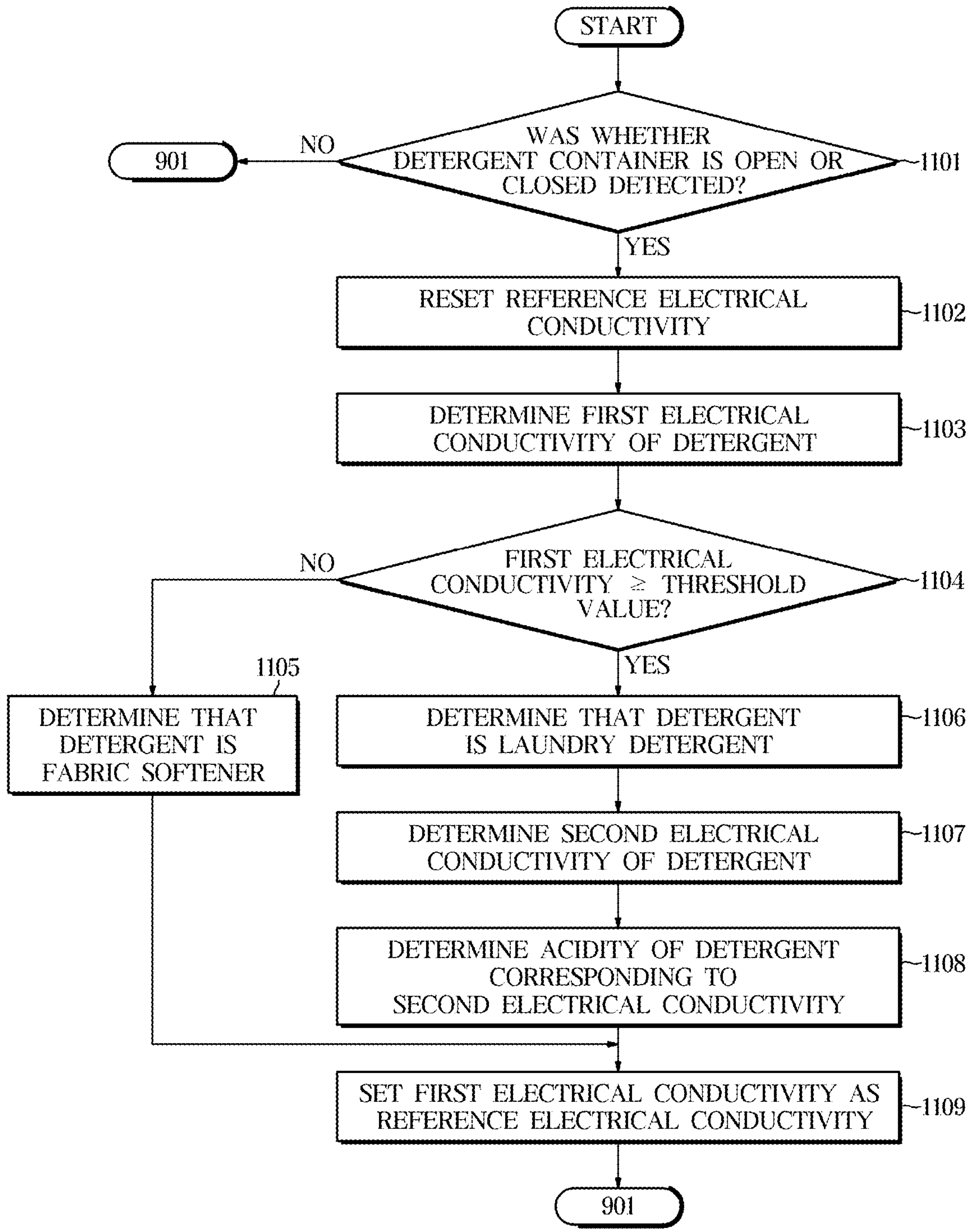
START
WAS WHETHER DETERGENT CONTAINER IS OPEN OR CLOSED DETECTED?
1101
NO
901
YES
RESET REFERENCE ELECTRICAL CONDUCTIVITY
1102
DETERMINE FIRST ELECTRICAL CONDUCTIVITY OF DETERGENT
1103
FIRST ELECTRICAL CONDUCTIVITY ≥ THRESHOLD VALUE?
1104
NO
YES
1105
DETERMINE THAT DETERGENT IS FABRIC SOFTENER
DETERMINE THAT DETERGENT IS LAUNDRY DETERGENT
1106
DETERMINE SECOND ELECTRICAL CONDUCTIVITY OF DETERGENT
1107
DETERMINE ACIDITY OF DETERGENT CORRESPONDING TO SECOND ELECTRICAL CONDUCTIVITY
1108
SET FIRST ELECTRICAL CONDUCTIVITY AS REFERENCE ELECTRICAL CONDUCTIVITY
1109
901

FIG. 12

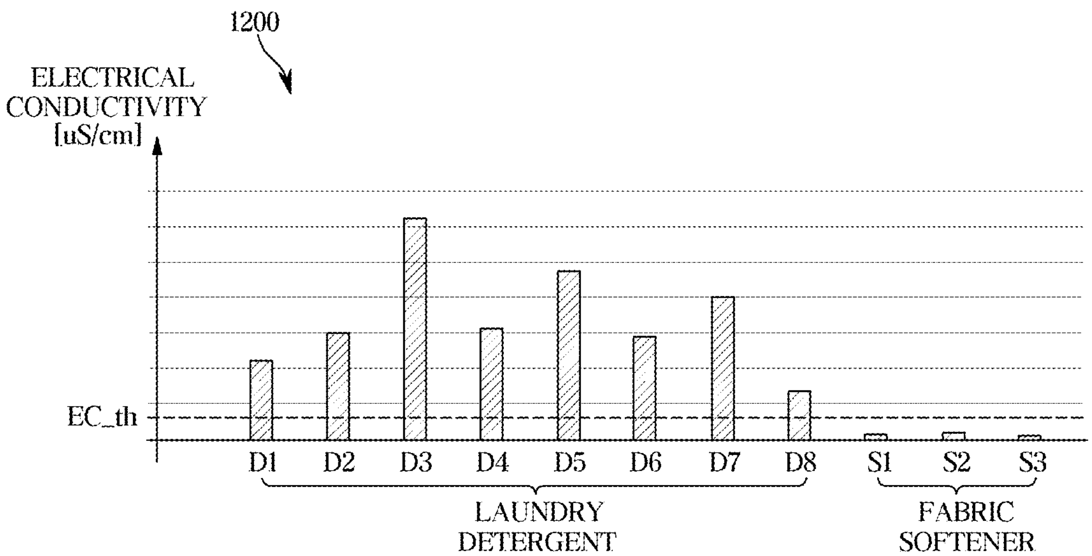
1200
ELECTRICAL
CONDUCTIVITY
[uS/cm]
EC_th
D1
D2
D3
D4
D5
D6
D7
D8
S1
S2
S3
LAUNDRY
DETERGENT
FABRIC
SOFTENER

FIG. 13

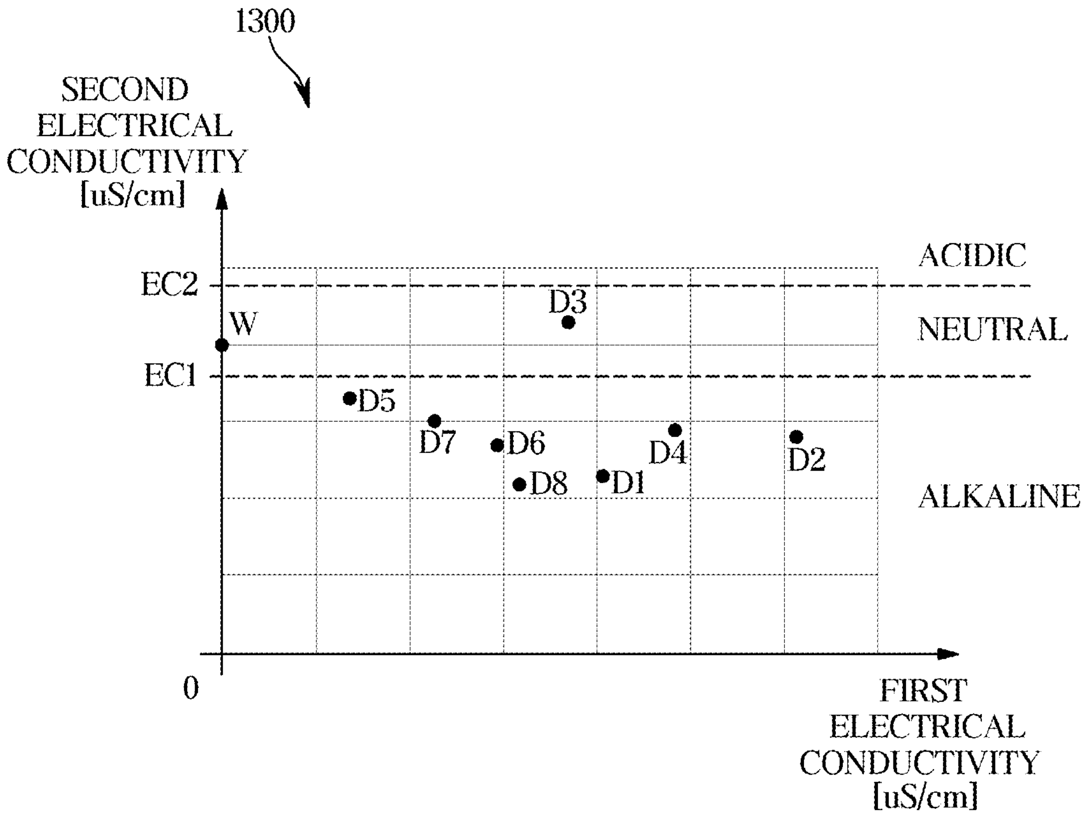
1300
SECOND ELECTRICAL CONDUCTIVITY [uS/cm]
EC2
EC1
W
D3
D5
D7
D6
D8
D1
D4
D2
ACIDIC
NEUTRAL
ALKALINE
0
FIRST ELECTRICAL CONDUCTIVITY [uS/cm]

FIG. 14

1400

	FIRST CHAMBER	SECOND CHAMBER	USER INTERFACE (DISPLAY)
Case1	LAUNDRY DETERGENT	FABRIC SOFTENER	- FIRST CHAMBER: LAUNDRY DETERGENT, ACIDITY - SECOND CHAMBER: FABRIC SOFTENER
Case2	LAUNDRY DETERGENT	LAUNDRY DETERGENT	- FIRST CHAMBER: LAUNDRY DETERGENT, ACIDITY - SECOND CHAMBER: LAUNDRY DETERGENT, ACIDITY CHAMBER SELECTION NOTIFICATION
Case3	FABRIC SOFTENER	LAUNDRY DETERGENT	- FIRST CHAMBER: FABRIC SOFTENER - SECOND CHAMBER: LAUNDRY DETERGENT, ACIDITY
Case4	FABRIC SOFTENER	FABRIC SOFTENER	- FIRST CHAMBER: FABRIC SOFTENER - FIRST CHAMBER: FABRIC SOFTENER - DETERGENT ERROR NOTIFICATION

WASHING MACHINE AND METHOD FOR CONTROLLING THE SAME

CROSS-REFERENCE TO RELATED APPLICATIONS

[0001] This is a continuation application, under 35 U.S.C. § 111(a), of International Application No. PCT/KR2025/002110, filed Feb. 13, 2025, which claims priority under 35 U.S.C. § 119 to Korean Patent Application No. 10-2024-0034146, filed Mar. 11, 2024, in the Korean Intellectual Property Office, the disclosures of which are incorporated herein in their entireties by reference.

TECHNICAL FIELD

[0002] The disclosure relates to a washing machine including a detergent supply device, and a method for controlling the same.

BACKGROUND ART

[0003] A washing machine is an apparatus for washing laundry through friction by stirring laundry, water, and a detergent together inside a tub with a driving force of a motor.

[0004] The washing machine includes a main body forming an exterior, and a detergent supply device in the main body. While water is supplied to the tub of the washing machine, the detergent supply device allows detergent to be evenly mixed and supplied together with the supplied water.

[0005] The detergent supply device may include a detergent container that accommodates detergent, and a detergent pump that pumps the detergent accommodated in the detergent container. The detergent container may accommodate various detergents, such as powdered detergent, liquid detergent, and fabric softener. The liquid detergent or fabric softener in the detergent container may be drawn into the detergent pump, and discharged to the outside of the detergent pump to move with the washing water to the tub.

[0006] Existing washing machines are unable to automatically measure a concentration and acidity of the detergent in the detergent container.

DISCLOSURE

[0007] The disclosure provides a washing machine that may detect changes in concentration of detergent in a detergent container and provide a detergent replenishment notification based on the change in detergent concentration, and a method for controlling the same.

[0008] The disclosure provides a washing machine that may adjust the amount of detergent to be supplied to a tub according to changes in concentration of detergent in a detergent container, and a method for controlling the same.

[0009] The disclosure provides a washing machine that may detect a concentration and acidity of detergent accommodated in each of a plurality of chambers forming a detergent container and automatically identify the type of detergent in each of the plurality of chambers, and a method for controlling the same.

[0010] In accordance with the present disclosure, a washing machine may include: a tub; a detergent container configured to accommodate detergent to be supplied to the tub; a plurality of electrodes in the detergent container; a user interface configured to provide information to a user; and a controller configured to, with detergent accommodated in the detergent container: apply a current having a defined frequency to electrodes of the plurality of electrodes, detect a voltage between electrodes of the plurality of electrodes, determine an electrical conductivity of the detergent based on the detected voltage, and provide, based on the determined electrical conductivity, information about a state of the detergent through the user interface and/or adjust an amount of the detergent to be supplied to the tub.

[0011] The controller may be further configured to, with the detergent accommodated in the detergent container, provide, based on a difference between the determined electrical conductivity and a reference electrical conductivity being greater than or equal to a defined reference value, a detergent replenishment notification through the user interface indicating that replenishment of the detergent is required.

[0012] The controller may be further configured to, with the detergent accommodated in the detergent container, adjust the amount of the detergent to be supplied to the tub based on a difference between the determined electrical conductivity and a reference electrical conductivity.

[0013] The controller may be further configured to, with the detergent accommodated in the detergent container, adjust the amount of the detergent to be supplied to the tub by adjusting a detergent supply time for supplying the detergent to the tub.

[0014] The controller may be further configured to, with the detergent accommodated in the detergent container: increase the detergent supply time to increase the amount of the detergent to be supplied to the tub, in response to the determined electrical conductivity being less than the reference electrical conductivity, and decrease the detergent supply time to decrease the amount of the detergent to be supplied to the tub, in response to the determined electrical conductivity being greater than the reference electrical conductivity.

[0015] The controller may be further configured to, with the detergent accommodated in the detergent container, provide, based on the determined electrical conductivity being increased by a defined limit value over a reference electrical conductivity, a detergent hardening notification through the user interface to warn that the detergent is hardened.

[0016] The controller may be further configured to: detect an opening and closing of the detergent container, and set the reference electrical conductivity based on the opening and closing of the detergent container being detected.

[0017] The controller may be further configured to: detect an opening and closing of the detergent container, and set the reference electrical conductivity to the electrical conductivity determined after the detergent container is opened and closed.

[0018] The defined frequency may be a first frequency, the current may be a first current, the detected voltage may be a first voltage, and the determined electrical conductivity may be a first electrical conductivity, and the controller may be further configured to, with the detergent accommodated in the detergent container: determine a concentration of the detergent based on the first electrical conductivity, apply a second current having a second frequency different from the first frequency to electrodes of the plurality of electrodes, detect a second voltage between electrodes of the plurality of electrodes, determine a second electrical conductivity of the detergent based on the detected second voltage, and determine an acidity of the detergent based on the determined second electrical conductivity.

[0019] The controller may be further configured to, with the detergent accommodated in the detergent container, determine whether the detergent is laundry detergent or fabric softener based on the determined concentration.

[0020] The detergent container may include a first chamber and a second chamber, and the controller may be further configured to, with the detergent accommodated in the first chamber and the second chamber, provide information through the user interface about whether the detergent accommodated in each of the first chamber and the second chamber is laundry detergent or fabric softener.

[0021] The controller may be further configured to provide a chamber selection notification through the user interface to guide a selection of one of the first chamber and the second chamber, in response to the laundry detergent being accommodated in both the first chamber and the second chamber.

[0022] The controller may be further configured to provide a detergent error notification through the user interface in response to the fabric softener being accommodated in both the first chamber and the second chamber.

[0023] In accordance with the present disclosure, in a method for controlling a washing machine including a tub, a detergent container configured to accommodate detergent to be supplied to the tub, a plurality of electrodes in the detergent container, and a user interface configured to provide information to a user, the method may include: with detergent accommodated in the detergent container, applying a current having a defined frequency to electrodes of the plurality of electrodes, detecting a voltage between electrodes of the plurality of electrodes, determining an electrical conductivity of the detergent based on the detected voltage, and providing, based on the determined electrical conductivity, information about a state of the detergent through the user interface and/or adjusting an amount of the detergent to be supplied to the tub.

[0024] The providing may include providing, based on a difference between the determined electrical conductivity and a reference electrical conductivity being greater than or equal to a defined reference value, a detergent replenishment notification indicating that replenishment of the detergent is required.

[0025] According to the disclosure, the washing machine and the method for controlling the same may detect changes in concentration of detergent in a detergent container and provide a detergent replenishment notification based on the change in detergent concentration. In addition, the washing machine and the method for controlling the same may adjust the amount of detergent to be supplied to a tub according to changes in concentration of detergent in a detergent container. Accordingly, the washing machine may minimize changes in washing performance due to the changes in detergent concentration.

[0026] According to the disclosure, the washing machine and the method for controlling the same may automatically identify the type of detergent in each of a plurality of chambers forming a detergent container and provide information about the detergent in each of the chambers. In addition, the washing machine and the method for controlling the same may automatically control operation of a detergent pump depending on the type of detergent accommodated in each of the chambers, and thus user convenience may be improved.

DESCRIPTION OF DRAWINGS

[0027] FIG. **1** illustrates a washing machine according to an embodiment.

[0028] FIG. **2** is a cross-sectional view of a washing machine according to an embodiment.

[0029] FIG. **3** is a view illustrating a detergent supply device of a washing machine according to an embodiment.

[0030] FIG. **4** is a view illustrating some separate components of a detergent supply device according to an embodiment.

[0031] FIG. **5** is a cross-sectional view of a detergent supply device according to an embodiment.

[0032] FIG. **6** is a view illustrating a floating device and detergent detection electrodes of a detergent supply device according to an embodiment.

[0033] FIG. **7** is a control block diagram of a washing machine according to an embodiment.

[0034] FIG. **8** is a flowchart briefly illustrating a method for controlling a washing machine according to an embodiment.

[0035] FIG. **9** is a flowchart illustrating the method described in FIG. **8** in greater detail.

[0036] FIG. **10** is a flowchart illustrating an embodiment of adjusting a detergent supply time described in FIG. **9** in greater detail.

[0037] FIG. **11** is a flowchart illustrating a method of determining a reference electrical conductivity and determining a type of detergent described in FIG. **9**.

[0038] FIG. **12** is a graph showing electrical conductivities corresponding to various detergent concentrations.

[0039] FIG. **13** is a graph showing first electrical conductivities corresponding to various detergent concentrations and second electrical conductivities corresponding to various detergent acidities.

[0040] FIG. **14** is a table illustrating example information provided for a detergent accommodated in a plurality of chambers of a detergent container.

MODES OF THE DISCLOSURE

[0041] Various embodiments of the disclosure and terms used herein are not intended to limit the technical features described herein to specific embodiments, and should be understood to include various modifications, equivalents, or substitutions of the corresponding embodiments.

[0042] In describing of the drawings, similar reference numerals may be used for similar or related elements.

[0043] The singular form of a noun corresponding to an item may include one or more of the items unless clearly indicated otherwise in a related context.

[0044] In the disclosure, phrases, such as "A or B", "at least one of A and B", "at least one of A or B", "A, B or C", "at least one of A, B and C", and "at least one of A, B, or C" may include any one or all possible combinations of the items listed together in the corresponding phrase among the phrases.

[0045] As used herein, the term "and/or" includes any and all combinations of one or more of the associated listed items.

[0046] Terms such as "1st", "2nd", "primary", or "secondary" may be used simply to distinguish an element from other elements, without limiting the element in other aspects (e.g., importance or order).

[0047] When an element (e.g., a first element) is referred to as being "(functionally or communicatively) coupled" or "connected" to another element (e.g., a second element), the first element may be connected to the second element, directly (e.g., wired), wirelessly, or through a third element.

[0048] It will be understood that when the terms "includes", "comprises", "including", and/or "comprising" are used in the disclosure, they specify the presence of the specified features, figures, steps, operations, components, members, or combinations thereof, but do not preclude the presence or addition of one or more other features, figures, steps, operations, components, members, or combinations thereof.

[0049] When a given element is referred to as being "connected to", "coupled to", "supported by" or "in contact with" another element, it is to be understood that it may be directly or indirectly connected to, coupled to, supported by, or in contact with the other element. When a given element is indirectly connected to, coupled to, supported by, or in contact with another element, it is to be understood that it may be connected to, coupled to, supported by, or in contact with the other element through a third element.

[0050] It will also be understood that when an element is referred to as being "on" another element, it may be directly on the other element or intervening elements may also be present.

[0051] A washing machine according to various embodiments may perform washing, rinsing, spin-drying, and drying processes. The washing machine is an example of a clothes treating apparatus, and the clothes treating apparatus is a concept including a device capable of washing clothes (objects to be washed, and objects to be dried), a device capable of drying clothes, and a device capable of washing and drying clothes.

[0052] The washing machine according to various embodiments may include a top-loading washing machine in which a laundry inlet for inserting or removing laundry is provided to face upward, or a front-loading washing machine in which a laundry inlet is provided to face forward. The washing machine according to various embodiments may include a washing machine of a loading type other than the top-loading washing machine and the front-loading washing machine.

[0053] For the top-loading washing machine, laundry may be washed using water current generated by a rotating body such as a pulsator. For the front-loading washing machine, laundry may be washed by repeatedly lifting and lowering laundry by rotating a drum. The front-loading washing machine may include a dryer combined washing machine capable of drying laundry stored in a drum. The dryer combined washing machine may include a hot air supply device for supplying high-temperature air into the drum and a condensing device for removing moisture from air discharged from the drum. For example, the dryer combined washing machine may include a heat pump device. The washing machine according to various embodiments may include a washing machine using a washing method other than the above-described washing method.

[0054] The washing machine according to various embodiments may include a main body accommodating various components therein. The main body may be provided in the form of a box including a laundry inlet on one side thereof.

[0055] The washing machine may include a door for opening and closing the laundry inlet. The door may be rotatably mounted to the main body by a hinge. At least a portion of the door may be transparent or translucent to allow the inside of the main body to be visible.

[0056] The washing machine may include a tub disposed within the main body to store water. The tub may be formed in a substantially cylindrical shape with a tub opening formed on one side thereof. The tub may be disposed inside the main body in such a way that the tub opening corresponds to the laundry inlet.

[0057] The tub may be connected to the main body by a damper. The damper may absorb vibration generated when the drum rotates, and the damper may reduce vibration transmitted to the main body.

[0058] The washing machine may include a drum provided to accommodate laundry.

[0059] The drum may be disposed inside the tub such that a drum opening provided on one side of the drum corresponds to the laundry inlet and the tub opening. Laundry may pass sequentially through the laundry inlet, the tub opening, and the drum opening and then be received in the drum or removed from the drum.

[0060] The drum may perform each operation according to washing, rinsing, and/or spin-drying while rotating in the tub. A plurality of through holes may be formed in a cylindrical wall of the drum to allow water stored in the tub to be introduced into or to be discharged from the drum.

[0061] The washing machine may include a driving device configured to rotate the drum. The driving device may include a drive motor and a rotating shaft for transmitting a driving force generated by the drive motor to the drum. The rotating shaft may penetrate the tub to be connected to the drum.

[0062] The driving device may perform respective operations according to washing, rinsing, and/or spin-drying, or drying processes by rotating the drum in a forward or reverse direction.

[0063] The washing machine may include a water supply device configured to supply water to the tub. The water supply device may include a water supply pipe and a water supply valve disposed in the water supply pipe. The water supply pipe may be connected to an external water supply source. The water supply pipe may extend from an external water supply source to a detergent supply device and/or the tub. Water may be supplied to the tub through the detergent supply device. Alternatively, water may be supplied to the tub without passing through the detergent supply device.

[0064] The water supply valve may open or close the water supply pipe in response to an electrical signal from a controller. The water supply valve may allow or block the supply of water to the tub from an external water supply source. The water supply valve may include a solenoid valve configured to open or close in response to an electrical signal.

[0065] The washing machine may include the detergent supply device configured to supply detergent to the tub. The detergent supply device may include a manual detergent supply device that requires a user to enter detergent to be used for each washing, and an automatic detergent supply device that stores a large amount of detergent and automatically adds a predetermined amount of detergent during washing. The detergent supply device may include a detergent container for storing detergent. The detergent supply device may be configured to supply detergent into the tub during a water supply process. Water supplied through the water supply pipe may be mixed with detergent via the detergent supply device. Water mixed with detergent may be supplied into the tub. Detergent is used as a term including detergent for pre-washing, detergent for main washing, fabric softener, bleach, etc., and the detergent container may be partitioned into a storage region for the pre-washing detergent, a storage region for the main washing detergent, a storage region for the fabric softener, and a storage region for the bleach.

[0066] The washing machine may include a drainage device configured to discharge water contained in the tub to the outside. The drainage device may include a drain pipe extending from a bottom of the tub to the outside of the main body, a drain valve disposed on the drain pipe to open or close the drain pipe, and a pump disposed on the drain pipe. The pump may pump water from the drain pipe to the outside of the main body.

[0067] The washing machine may include a control panel disposed on one side of the main body. The control panel may provide a user interface for interaction between a user and the washing machine. The user interface may include at least one input interface and at least one output interface.

[0068] The at least one input interface may convert sensory information received from a user into an electrical signal.

[0069] The at least one input interface may include a power button, an operation button, a course selection dial (or a course selection button), and a washing/rinsing/spin-drying setting button. The at least one input interface may include a tact switch, a push switch, a slide switch, a toggle switch, a micro switch, a touch switch, a touch pad, a touch screen, a jog dial, and/or a microphone.

[0070] The at least one output interface may visually or audibly transmit information related to the operation of the washing machine to a user.

[0071] For example, the at least one output interface may transmit information related to a washing course, operation time of the washing machine, and washing/rinsing/spin-drying settings to the user. Information about the operation of the washing machine may be output via a screen, an indicator, or a voice. The at least one output interface may include a liquid crystal display (LCD) panel, a light emitting diode (LED) panel, or a speaker.

[0072] The washing machine may include a communication module for wired and/or wireless communication with an external device.

[0073] The communication module may include at least one of a short-range wireless communication module and a long-range wireless communication module.

[0074] The communication module may transmit data to an external device (e.g., a server, a user device, and/or a home appliance) or receive data from the external device. For example, the communication module may establish communication with a server and/or a user device and/or a home appliance, and transmit and receive various types of data.

[0075] For the communication, the communication module may establish a direct (e.g., wired) communication channel or a wireless communication channel between the external devices, and support the performance of the communication through the established communication channel. According to an embodiment, the communication module may include a wireless communication module (e.g., a cellular communication module, a short-range wireless communication module, or a global navigation satellite system (GNSS) communication module) or a wired communication module (e.g., a local area network (LAN) communication module, or a power line communication module). Among these communication modules, the corresponding communication module may communicate with an external device through a first network (e.g., a short-range wireless communication network such as Bluetooth, wireless fidelity (Wi-Fi) direct, or infrared data association (IrDA)) or a second network (e.g., a long-range wireless communication network such as a legacy cellular network, a 5G network, a next-generation communication network, the Internet, or a computer network (e.g., LAN or wide area network WAN)). These various types of communication modules may be integrated as a single component (e.g., a single chip) or implemented as a plurality of separate components (e.g., multiple chips).

[0076] The short-range wireless communication module may include a Bluetooth communication module, a Bluetooth Low Energy (BLE) communication module, a near field communication module, a WLAN (Wi-Fi) communication module, and a Zigbee communication module, an IrDA communication module, a Wi-Fi Direct (WFD) communication module, an ultrawideband (UWB) communication module, an Ant+ communication module, a microwave (uWave) communication module, etc., but is not limited thereto.

[0077] The long-range wireless communication module may include a communication module that performs various types of long-range wireless communication, and may include a mobile communication circuitry. The mobile communication circuitry transmits and receives radio signals with at least one of a base station, an external terminal, and a server in a mobile communication network.

[0078] According to an embodiment, the communication module may communicate with an external device such as a server, a user device and other home appliances through an access point (AP). The AP may connect a LAN, to which a washing machine or a user device is connected, to a WAN to which a server is connected. The washing machine or the user device may be connected to the server via the WAN. The controller may control various components of the washing machine (e.g., the drive motor, and the water supply valve). The controller may control various components of the washing machine to perform at least one operation including water supply, washing, rinsing, and/or spin-drying according to a user input. For example, the controller may control the drive motor to adjust the rotational speed of the drum or control the water supply valve of the water supply device to supply water to the tub.

[0079] The controller may include hardware such as a CPU or memory, and software such as a control program. For example, the controller may include at least one memory for storing an algorithm and program-type data for controlling the operation of components in the washing machine, and at least one processor configured to perform the above-mentioned operation by using the data stored in the at least one memory. The memory and the processor may each be implemented as separate chips. The processor may include one or more processor chips or may include one or more processing cores. The memory may include one or more memory chips or one or more memory blocks. Alternatively, the memory and the processor may be implemented as a single chip.

[0080] The terms "front," "rear," "side," "left," "right," "upper," "lower", and the like are defined with reference to the drawings, and are not intended to limit the shape and position of any element. For example, the terms "upper," and "lower," may be defined based on the Z direction shown in the drawings. The terms "front," and "rear," may be defined based on the X direction shown in the drawings. The terms "left," and "right," may be defined based on the Y direction shown in the drawings.

[0081] Hereinafter, various embodiments according to the disclosure will be described in detail with reference to the accompanying drawings.

[0082] FIG. **1** illustrates a washing machine according to an embodiment. FIG. **2** is a cross-sectional view of a washing machine according to an embodiment.

[0083] Referring to FIG. **1** and FIG. **2**, a washing machine **1** according to an embodiment may include a main body **10** accommodating various components therein. The main body **10** may form an exterior of the washing machine **1**. A laundry inlet **11** may be formed on one side of the main body **10**. The main body **10** may be in the form of a box including the laundry inlet **11** on one side thereof. The laundry inlet **11** may be formed on a front side of the main body **10**.

[0084] The washing machine **1** may include a user interface **15** for interaction between a user and the washing machine **1**. The user interface **15** may include an input interface **15***a* for obtaining a user input. The user interface **15** may include an output interface **15***b* for outputting information about an operation of the washing machine **1**. The input interface **15***a* may include various buttons and/or dials. The output interface **15***b* may include a display.

[0085] The user interface **15** may be provided at various locations of the main body **10**. For example, the user interface **15** may be provided on the front side of the main body **10**. The user interface **15** may also be provided on an upper side of the main body **10**.

[0086] The main body **10** may include a detergent container panel **16**. The detergent container panel **16** may be located on an upper front side of the main body **10**. The detergent container panel **16** may be coupled to a detergent container **150** to be described below. The detergent container panel **16** may be coupled to a front side of the detergent container **150**. The detergent container panel **16** may be grasped by a user. The user may hold the detergent container panel **16** and take the detergent container **150** out of the main body **10** or insert the detergent container **150** into the main body **10**.

[0087] The washing machine **1** may include a door **41** to open or close the laundry inlet **11**. The door **41** may be rotatably coupled to the main body **10** to open or close the laundry inlet **11**. The door **41** may be rotatably coupled to the main body **10** by a hinge **42**. The door **41** may be coupled to the front side of the main body **10**.

[0088] The washing machine **1** may include a tub **20** disposed inside the main body **10**. The tub **20** may store water. A tub opening **21** may be formed on one side of the tub **20**. The tub opening **21** may be disposed to correspond to the laundry inlet **11**. For example, the tub **20** may be formed in a cylindrical shape with the tub opening **21** formed on approximately one side thereof.

[0089] The tub **20** may be connected to a damper **29**. The tub **20** may be connected to the main body **10** by the damper **29**. The tub **20** may be supported by the damper **29**. The damper **29** may absorb vibration generated during rotation of a drum **30**, and reduce vibration transmitted to the main body **10**.

[0090] The washing machine **1** may include the drum **30** disposed inside the tub **20**. The drum **30** may accommodate laundry. The drum **30** may be rotatable inside the tub **20**. The drum **30** may rotate during each of a washing process, a rinsing process, and a spin-drying process. A rotation speed and rotation direction of the drum **30** may be adjusted differently for each process.

[0091] A drum opening **31** may be formed on one side of the drum **30**. The drum opening **31** may be disposed to correspond to the laundry inlet **11** and the tub opening **21**. Laundry may pass sequentially through the laundry inlet **11**, the tub opening **21**, and the drum opening **31** and then be received in the drum **30**. Conversely, laundry may pass sequentially through the drum opening **31**, the tub opening **21**, and the laundry inlet **11**, and then taken out from the drum **30**.

[0092] The drum **30** may include a plurality of through holes **34**. Water stored in the tub **20** may be introduced into the drum **30** through the through holes **34**, or water in the drum **30** may be discharged to the tub **20** through the through holes **34**.

[0093] At least one lifter **35** that lifts laundry when the drum **30** rotates may be installed on an inner surface of the drum **30**.

[0094] The washing machine **1** may include a driving device **60** configured to rotate the drum **30**. The driving device **60** may include a drive motor generating a driving force for rotating the drum **30** and a rotating shaft for transmitting the driving force generated by the drive motor to the drum **30**. The rotating shaft may penetrate the tub **20** to be connected to the drum **30**. The driving device **60** may rotate the drum **30** forward or backward.

[0095] The washing machine **1** may include a water supply device **50** to supply water to the tub **20**. The water supply device **50** may include a water supply pipe **51** connected to an external water supply source, and a water supply valve **52** for opening and closing the water supply pipe **51**. The water supply pipe **51** may be disposed in an upper part in the main body **10**.

[0096] The water supply pipe **51** may extend from the external water supply source to a detergent supply device **100** to be described below, and water may be supplied to the tub **20** from the external water supply source through the detergent supply device **100**. Alternatively, the water supply pipe **51** may extend to the tub **20** from the external water supply source. In this case, water may be supplied to the tub **20** without passing through the detergent supply device **100**.

[0097] The washing machine **1** may include the detergent supply device **100**. The detergent supply device **100** may supply detergent to the tub **20**. The detergent supply device **100** may be configured to supply water mixed with detergent to the tub **20**.

[0098] The detergent supply device **100** may include a detergent container **150** (see FIG. **3**) for storing detergent. The detergent supply device **100** may supply detergent accommodated in the detergent container **150** to the inside of the tub **20**. Water supplied through the water supply pipe

51 may pass through the detergent supply device **100** and be mixed with detergent. The water mixed with detergent may be supplied to the tub **20**.

[0099] The detergent supply device **100** may be connected to the tub **20** through a connecting pipe **53**. The water mixed with detergent in the detergent supply device **100** may be supplied to the tub **20** through the connecting pipe **53**. The detergent supply device **100** may be disposed in the upper part in the main body **10**. For example, the detergent supply device **100** may be disposed above the tub **20**.

[0100] The washing machine **1** may include a drainage device **54** to discharge water in the tub **20** to the outside. The drainage device **54** may include a drain pipe **55** extending from a bottom of the tub **20** to the outside of the main body **10**, a drain valve **56** for opening and closing the drain pipe **55**, and a drain pump **57** disposed on the drain pipe **55**.

[0101] The configuration of the washing machine **1** is not limited to those described above. The washing machine **1** may further include various components in addition to the above-described components.

[0102] In addition, the washing machine **1** is described as a front-loading type, but is not limited thereto. A method for controlling the washing machine **1** according to the disclosure may also be applied to a top-loading washing machine and a dryer combined washing machine.

[0103] FIG. **3** is a view illustrating a detergent supply device of a washing machine according to an embodiment.

[0104] Referring to FIG. **3**, the detergent supply device **100** of the washing machine **1** according to an embodiment may include the detergent container **150**. The detergent container **150** may accommodate detergent to be supplied to the tub **20**.

[0105] The detergent container **150** may include at least one chamber for accommodating detergent. Various types of detergents, such as detergent for pre-washing, detergent for main washing, fabric softener, and bleach, may be accommodated in the chambers of the detergent container **150**.

[0106] For example, the detergent container **150** may include a first chamber **151**, a second chamber **152**, and a third chamber **153**. Liquid detergent may be accommodated in the first chamber **151** and the second chamber **152**. For example, liquid laundry detergent or liquid fabric softener may be accommodated in the first chamber **151** and the second chamber **152**. The third chamber **153** may accommodate powdered detergent.

[0107] The types of detergent that may be accommodated in each of the first chamber **151**, the second chamber **152**, and the third chamber **153** are not limited to the above examples.

[0108] The first chamber **151**, the second chamber **152**, and the third chamber **153** may be partitioned by a chamber partition **154**. The chamber partition **154** may be formed to partition the first chamber **151** and the second chamber **152** inside the detergent container **150**. For example, the chamber partition **154** may extend upward from a bottom side **150***a* of the detergent container **150**.

[0109] Although it is shown in FIG. **3** that the first chamber **151** is formed on the right side of the detergent container **150** and the second chamber **152** is formed on the left side of the detergent container **150**, the positions of the first chamber **151** and the second chamber **152** are not limited thereto. In addition, it is shown in FIG. **3** that the third chamber **153** is positioned between the first chamber **151** and the second chamber **152**, but the position of the third chamber **153** is also not limited thereto.

[0110] The detergent container **150** may be movable relative to the main body **10**. As described above, the detergent container **150** may be taken out from the main body **10** or inserted into the main body **10**. The detergent container **150** may be mounted to be movable relative to a housing **140**. The detergent container **150** may be mounted to the housing **140** such that the detergent container **150** may be inserted into the housing **140** or withdrawn from the housing **140**.

[0111] One side of the detergent container **150** may be open. For example, an upper side of the detergent container **150** may be open. Upper sides of the first chamber **151**, the second chamber **152**, and the third chamber **153** may be open to allow detergent to be added to each chamber.

[0112] The detergent supply device **100** may include a detergent container cover **130** to cover the detergent container **150**. The detergent container cover **130** may be disposed on an upper part of the detergent container **150**. The detergent container cover **130** may be detachably coupled to the upper part of the detergent container **150**.

[0113] The detergent container cover **130** may cover the first chamber **151**. The detergent container cover **130** may cover the second chamber **152**. However, the third chamber **153** may not be covered by the detergent container cover **130**.

[0114] The detergent container cover **130** may have a detergent inlet formed to allow detergent to be introduced into the detergent container **150**. The detergent inlet may be opened and closed by a detergent container cap **131**. The detergent container cap **131** may be detachably coupled to the detergent container cover **130**. A plurality of detergent container caps **131** may be provided to open and close each of the first chamber **151** and the second chamber **152**.

[0115] Hereinafter, a configuration in which detergent in the detergent container **150** is discharged and the detergent discharged from the detergent container **150** is mixed with water and discharged to the tub **20** is described in detail with an example.

[0116] The detergent supply device **100** may include a water supply cover **110**. The water supply cover **110** may form the exterior of the detergent supply device **100** together with the housing **140**. The water supply cover **110** may be coupled to an upper part of a water supply plate **120**. The water supply cover **110** may be coupled to the water supply plate **120** to seal an upper side of the water supply plate **120**. The water supply cover **110** may be coupled to the water supply plate **120** to form a flow path through which washing water supplied from the water supply pipe **51** moves.

[0117] The water supply cover **110** may include a washing water inlet **111** connected to the water supply pipe **51**. Water supplied from the water supply pipe **51** may be introduced into the detergent supply device **100** through the washing water inlet **111**.

[0118] The detergent supply device **100** may include the water supply plate **120** coupled to the water supply cover **110**. The water supply plate **120** may be positioned between the water supply cover **110** and the housing **140**. The water supply plate **120** may be positioned between the water supply cover **110** and the detergent container **150**. The water supply plate **120** may be positioned between the water supply cover **110** and the detergent container cover **130**. For example, the water supply plate **120** may be disposed above the detergent container cover **130**. The water supply plate

120 may include a guide partition wall **121** defining a flow path to allow water to flow. Together with the water supply cover **110**, the water supply plate **120** may form a flow path through which the water supplied from the water supply pipe **51** moves.

[0119] The water supply plate **120** may include first washing water flow holes **122** formed to allow water flowing into a space between the water supply cover **110** and the water supply plate **120** from the water supply pipe **51** to be discharged to the third chamber **153**. The first washing water flow holes **122** may be formed by penetrating the water supply plate **120**. The first washing water flow holes **122** may be formed in a region of the water supply plate **120** facing the third chamber **153**.

[0120] The water supply plate **120** may include a second washing water flow hole **123** formed to allow a portion of the water supplied through the water supply pipe **51** to flow directly into the housing **140** (i.e., a mixing chamber **141**). The second washing water flow hole **123** may be formed upstream of the first washing water flow holes **122**. The second washing water flow hole **123** may be formed adjacent to the washing water inlet **111**.

[0121] The detergent supply device **100** may include the housing **140** forming the exterior of the detergent supply device **100** together with the water supply cover **110**. The housing **140** may be mounted to the main body **10**. The housing **140** may be fixed to the main body **10**.

[0122] The housing **140** may cover the detergent container **150**. The housing **140** may movably support the detergent container **150**. An opening may be formed in a front side of the housing **140**. The detergent container **150** may be withdrawn from the housing **140** or inserted into the housing **140** through the opening on the front side of the housing **140**.

[0123] The housing **140** may include the mixing chamber **141** formed inside the housing **140**. Liquid detergent in the first chamber **151** or the second chamber **152** may be moved to the mixing chamber **141** by a detergent pump **400** and then supplied to the tub **20**. The third chamber **153** may be connected to the mixing chamber **141** through a flue opening **153*a*** (see FIG. **4**). Powdered detergent accommodated in the third chamber **153** may be mixed with water introduced through the first washing water flow holes **122** in the third chamber **153**. The water mixed with the powdered detergent may be moved to the mixing chamber **141** through the flue opening **153*a***.

[0124] An outlet **142** may be formed at the bottom of the housing **140** to discharge the detergent water from the mixing chamber **141** to the outside of the detergent supply device **100**. The outlet **142** may be connected to the connecting pipe **53**. The outlet **142** may be connected to the tub **20** through the connecting pipe **53**. The detergent water in the mixing chamber **141** may be discharged through the outlet **142** and then supplied to the tub **20** through the connecting pipe **53**.

[0125] The detergent supply device **100** may include the detergent pump **400**. The detergent pump **400** may move detergent accommodated in the detergent container **150** to the mixing chamber **141**. The detergent pump **400** may draw in detergent from the detergent container **150** and discharge the detergent to the mixing chamber **141**. For example, the detergent pump **400** may draw in detergent from the first chamber **151** or the second chamber **152** and discharge the detergent to the mixing chamber **141**.

[0126] The detergent pump **400** may be mounted behind the detergent container **150**. The detergent pump **400** may penetrate a pump opening **143** of the housing **140**. The detergent pump **400** may include a detergent intake portion **410** to draw in detergent from the detergent container **150**. A portion of the detergent intake portion **410** may penetrate a rear side of the detergent container **150** and be inserted into the detergent container **150**. A plurality of detergent intake portions **410** may be provided. For example, the detergent intake portions **410** may be provided corresponding to each of the first chamber **151** and the second chamber **152**.

[0127] The detergent pump **400** may include a detergent outlet **420** to discharge the detergent drawn in from the detergent container **150** into the mixing chamber **141**. The detergent outlet **420** may penetrate a rear side of the housing **140**. The detergent outlet **420** may be formed to penetrate the pump opening **143**.

[0128] A drive motor (not shown) generating a driving force may be provided in the detergent pump **400**. The drive motor of the detergent pump **400** may be controlled by a controller **500** of the washing machine **1**.

[0129] FIG. **4** is a view illustrating some separate components of a detergent supply device according to an embodiment. FIG. **5** is a cross-sectional view of a detergent supply device according to an embodiment. FIG. **6** is a view illustrating a floating device and detergent detection electrodes of a detergent supply device according to an embodiment.

[0130] Referring to FIG. **4**, FIG. **5** and FIG. **6**, the detergent supply device **100** may include a detergent amount detection device configured to detect the amount of detergent accommodated in the detergent container **150**. For example, the detergent amount detection device may include a floating device **200**, and a magnetic field sensor **300** to detect a magnetic field generated by the floating device **200**.

[0131] The floating device **200** may be disposed in the detergent container **150**. The floating device **200** may be movable in the detergent container **150**. A plurality of floating devices **200** may be provided. For example, the floating devices **200** may be disposed in each of the first chamber **151** and the second chamber **152**.

[0132] In a case where a defined amount of liquid detergent or more is accommodated in the detergent container **150**, the floating device **200** may float in the detergent container **150** due to the buoyancy of the liquid detergent. In a case where the amount of liquid detergent in the detergent container **150** is less than the defined amount, the floating device **200** may be positioned adjacent to or in contact with the bottom side **150*a*** of the detergent container **150**.

[0133] In a case where the defined amount of liquid detergent or more is accommodated in the detergent container **150**, the floating device **200** may rise from the bottom side **150*a*** of the detergent container **150** due to the buoyancy of the liquid detergent. As the amount of liquid detergent increases, the floating device **200** may rise higher from the bottom side **150*a*** of the detergent container **150**. Conversely, as the amount of liquid detergent in the detergent container **150** decreases, the floating device **200** may descend toward the bottom side **150*a*** of the detergent container **150**. That is, the floating device **200** may be movable in the vertical direction Z according to the amount of liquid detergent in the detergent container **150**.

[0134] The magnetic field sensor **300** may include various types of magnetic field sensors. As the floating device **200**

rises or descends, an intensity of the magnetic field detected by the magnetic field sensor **300** may change. The washing machine **1** may use the change in the intensity of the magnetic field detected by the magnetic field sensor **300** to identify the amount of liquid detergent accommodated in the detergent container **150**. A plurality of magnetic field sensors **300** may be provided to correspond to a plurality of floating devices **200**.

[0135] In a case where the floating device **200** is located at the bottom, the intensity of the magnetic field detected by the magnetic field sensor **300** may be the largest. In a case where the floating device **200** rises to a floating position due to the liquid detergent, the intensity of the magnetic field detected by the magnetic field sensor **300** may decrease because a distance between the floating device **200** and the magnetic field sensor **300** increases.

[0136] The magnetic field sensor **300** may be disposed on one side of the floating device **200**. For example, the magnetic field sensor **300** may be disposed below the floating device **200**. As shown in FIG. **5**, the magnetic field sensor **300** may be disposed lower than the detergent container **150**. The magnetic field sensor **300** may be mounted on a lower side of the housing **140**. Alternatively, the magnetic field sensor **300** may be disposed between the detergent container **150** and the housing **140**. The magnetic field sensor **300** may be mounted on a lower side of the detergent container **150**. The magnetic field sensor **300** may be disposed inside the detergent container **150**. The magnetic field sensor **300** may be disposed in various locations to efficiently detect a magnetic field generated by the floating device **200**.

[0137] The floating device **200** and the magnetic field sensor **300** disposed in the first chamber **151** may detect the amount of liquid detergent accommodated in the first chamber **151**. The floating device **200** and the magnetic field sensor **300** disposed in the second chamber **152** may detect the amount of liquid detergent accommodated in the second chamber **152**.

[0138] The bottom side **150*a*** of the detergent container **150** may be formed to slope downward toward the rear of the detergent container **150**. As the bottom side **150*a*** of the detergent container **150** slopes downward toward the rear of the detergent container **150**, the liquid detergent may flow toward the rear within the detergent container **150**. Because the detergent pump **400** is located at the rear side of the detergent container **150**, the detergent in the detergent container **150** may be drawn in more efficiently.

[0139] The floating device **200** may also be disposed in a rear part of the detergent container **150**. The floating device **200** may be disposed adjacent to the detergent pump **400** within the detergent container **150**. Accordingly, the amount of liquid detergent in the detergent container **150** may be detected more efficiently and accurately. For example, the floating device **200** accommodated in the first chamber **151** may be disposed adjacent to a rear end of the first chamber **151**. The floating device **200** accommodated in the second chamber **152** may be disposed so as to be spaced apart from a rear end of the second chamber **152** by a predetermined distance.

[0140] The detergent container **150** may include a guide rib **155** to guide the movement of the floating device **200**. The guide rib **155** may support the movement of the floating device **200** in the vertical direction Z. That is, the guide rib **155** may guide the floating device **200** to move between a bottom position and a floating position.

[0141] The guide rib **155** may be provided along a periphery of the floating case **220**. The guide rib **155** may form a movement space **155*s*** in which the floating device **200** may move in the vertical direction Z. The movement space **155*s*** may extend approximately in the vertical direction Z. The movement space **155*s*** may extend upward from the bottom side **150*a*** of the detergent container **150**.

[0142] The guide rib **155** may extend in the vertical direction Z. The guide rib **155** may extend upward from the bottom side **150*a*** of the detergent container **150**. The guide rib **155** may be formed integrally with the bottom side **150*a*** of the detergent container **150**. The guide rib **155** may extend from the bottom side **150*a*** of the detergent container **150** to a height approximately equal to the top of the detergent container **150**.

[0143] A plurality of guide ribs **155** may be provided. The plurality of guide ribs **155** may be arranged along the periphery of the floating case **220**. The plurality of guide ribs **155** may be spaced apart from each other or may be connected to each other.

[0144] The detergent container **150** may be further provided with a cap **156** that covers the movement space **155*s***. The cap **156** may be coupled to an upper end **155*a*** of each of the plurality of guide ribs **155**. The cap **156** may fix the upper end **155*a*** of each of the plurality of guide ribs **155**, thereby preventing the upper ends **155*a*** of the guide ribs **155** from moving out of the design position, such as expanding outwardly from the movement space **155*s*** or contracting inwardly of the movement space **155*s***. The cap **156** may be positioned at a height approximately equal to the top of the detergent container **150**.

[0145] Referring to FIG. **6**, the washing machine **1** may include a detergent detection electrode **430** to detect detergent in the detergent container **150**. A plurality of detergent detection electrodes **430** may be provided. For example, a pair of detergent detection electrodes **430** may be provided. The pair of detergent detection electrodes **430** may be positioned at the same height. The detergent detection electrode **430** may include a terminal portion that is electrically conductive. The terminal portion of the detergent detection electrode **430** may be positioned in the detergent container **150**.

[0146] Although it is described that the detergent detection electrode **430** is included in the detergent pump **400**, the disclosure is not limited thereto. The detergent detection electrode **430** may be provided separately from the detergent pump **400**.

[0147] The washing machine **1** may identify whether detergent is present in the detergent container **150** by using the detergent detection electrodes **430**. For example, in a case where the liquid detergent in the detergent container **150** contacts the detergent detection electrodes **430**, a current is applied to the detergent detection electrodes **430**, and thus a voltage may be detected between a pair of detergent detection electrodes **430**. As the voltage is detected between the plurality of detergent detection electrodes **430**, the washing machine **1** may identify that liquid detergent is accommodated in the detergent container **150**. Conversely, in a case where no voltage is detected between the plurality of detergent detection electrodes **430**, the washing machine **1** may identify that no liquid detergent is in the detergent container **150** or that there is only a small amount of liquid detergent that does not contact the detergent detection electrodes **430**.

[0148] The washing machine **1** may determine that liquid detergent is accommodated at or above a predetermined height in the detergent container **150**, because the voltage is detected between the plurality of detergent detection electrodes **430**. The predetermined height may represent the height from the bottom side **150***a* of the detergent container **150** to the detergent detection electrodes **430**. Conversely, based on no voltage being detected after applying current to the detergent detection electrodes **430**, the washing machine **1** may determine that very little liquid detergent is present in the detergent container **150** or that the liquid detergent is accommodated below the predetermined height.

[0149] The washing machine **1** may also apply a voltage between two detergent detection electrodes **430** and then detect the current flowing through the detergent detection electrodes **430**, thereby identifying whether detergent is present in the detergent container **150**.

[0150] In addition, the washing machine **1** may determine an electrical conductivity of the detergent based on the current applied to the plurality of detergent detection electrodes **430** and the detected voltage. The washing machine **1** may determine a concentration and/or an acidity of the detergent corresponding to the electrical conductivity of the detergent. Electrical conductivity is a unique physical property that indicates how well a material allows a flow of electricity. The reciprocal of electrical conductivity corresponds to impedance.

[0151] FIG. **7** is a control block diagram of a washing machine according to an embodiment.

[0152] Referring to FIG. **7**, the washing machine **1** may include the controller **500**. The controller **500** may be electrically connected to various components and/or devices of the washing machine **1**, and may control various components and/or devices. For example, the controller **500** may control the user interface **15**, the water supply valve **52**, the drain valve **56**, the drain pump **57**, the driving device **60**, and the detergent pump **400**. The controller **500** may be electrically connected to a detergent container open/close sensor **90**, a communication interface **80**, the magnetic field sensor **300**, and the detergent detection electrode **430**.

[0153] The controller **500** may include a plurality of control circuits, or may be an integrated single control circuit. The controller **500** may include a processor **510** and a memory **520**. The memory **520** may include volatile memory (e.g., static random access memory (S-RAM), dynamic random access memory (D-RAM)) and non-volatile memory (e.g., read only memory (ROM), an erasable programmable read only memory (EPROM)). The processor **510** and the memory **520** may be implemented as separate chips or as a single chip. In addition, a plurality of processors and a plurality of memories may be provided. The processor **510** may process various data and signals using instructions, data, programs, and/or software stored in the memory **520**. The processor **510** may include a single or a plurality of cores. The processor **510** may generate control signals for controlling components of the washing machine **1**.

[0154] The user interface **15** may obtain various user inputs and may output various information about the operation of the washing machine **1**. The user interface **15** may include the input interface **15***a* and the output interface **15***b*.

[0155] The controller **500** may control the operation of the washing machine **1** based on user input obtained through the user interface **15**. For example, the controller **500** may turn the washing machine **1** on or off based on user input for turning the washing machine **1** on or off. The controller **500** may determine a washing course of the washing machine **1** based on user input for selecting the washing course of the washing machine **1**. The controller **500** may start or stop a washing process, a rinsing process, and/or a spin-drying process based on user input through the start and/or stop buttons.

[0156] A variety of washing courses of the washing machine **1** may be provided. The various washing courses may be provided according to the type (e.g., bedding, underwear, etc.) and materials (e.g., cotton, wool, etc.) of laundry. For example, the washing course may include at least one of a standard washing course, intense washing course, delicate clothes washing course, bedding washing course, baby clothes washing course, towel washing course, boiling washing course, and/or outdoor clothes washing course.

[0157] Each of the washing courses may include different washing settings (e.g., washing temperature, a number of rinses, spin speed, etc.). According to a selection of one of the washing courses through the user interface **15** or an external user device, the controller **500** may control the washing machine **1** to perform a washing process, a rinsing process, and a spin-drying process according to the selected washing course. In addition, the washing course may include a rinsing and spin-drying course excluding a washing process, a rinsing course, and a spin-drying course. The washing courses are not limited to the above examples.

[0158] The controller **500** may control the user interface **15** to output various information about the operation of the washing machine **1**. For example, the user interface **15** may visually and/or audibly output information about the washing course, operation time, washing settings, rinsing settings, and/or drying settings of the washing machine **1**. In addition, the user interface **15** may output information about a state of detergent accommodated in the detergent container **150**.

[0159] The driving device **60** may rotate the drum **30** under the control of the controller **500**. The driving device **60** may include a drive motor. The controller **500** may control the drive motor to adjust a rotation speed of the drum **30**.

[0160] The controller **500** may control the opening and closing of the water supply valve **52**. The controller **500** may adjust an opening degree of the water supply valve **52**. The water supply valve **52** may open or close the water supply pipe **51** based on an electrical signal transmitted from the controller **500**.

[0161] The controller **500** may control the opening and closing of the drain valve **56**. The controller **500** may adjust an opening degree of the drain valve **56**. The drain valve **56** may open or close the drain pipe **55** based on an electrical signal transmitted from the controller **500**.

[0162] The drain pump **57** may discharge water in the tub **20** to the outside of the main body **10**. The controller **500** may control the drain pump **57** to allow water in the tub **20** to be discharged to the outside through the drain pipe **55**.

[0163] The communication interface **80** may include various communication circuitry for performing a wired communication and/or wireless communication with an external device (e.g., servers, user devices, and/or other home appliances). A user device may include various electronic devices, such as smartphones, laptops, smart watches, stationary type tablets, and speakers. A user input may be obtained not only through the user interface **15** but also through a user device.

[0164] The communication interface **80** may include at least one of short-range communication circuitry or long-range communication circuitry. The communication interface **80** may transmit data to or receive data from an external device. For example, the communication interface **80** may support cellular communication, wireless local area network (WLAN), home radio frequency (HomeRF), infrared communication, ultra-wide band (UWB) communication, Wi-Fi, Wi-Fi Direct, Bluetooth, and ad-hoc and/or Zigbee. The communication technologies supported by the communication interface **80** are not limited to the above.

[0165] The communication interface **80** may communicate with an external device through an access point (AP). The AP may connect a local area network (LAN) to which the washing machine **1** is connected to a wide area network (WAN) to which a server is connected. The washing machine **1** may be connected to the server through the WAN.

[0166] The detergent container open/close sensor **90** may detect the opening and closing of the detergent container **150**. The detergent container open/close sensor **90** may detect whether the detergent container **150** is inserted into the main body **10** or taken out from the main body **10**. The detergent container open/close sensor **90** may transmit an electrical signal corresponding to the opening and closing of the detergent container **150** to the controller **500**. The controller **500** may determine the opening and closing of the detergent container **150** based on the electrical signal transmitted from the detergent container open/close sensor **90**.

[0167] The magnetic field sensor **300** may detect a magnetic field generated by the floating device **200**. The magnetic field sensor **300** may transmit an electrical signal corresponding to the magnetic field of the floating device **200** to the controller **500**. The controller **500** may determine the amount of liquid detergent in the detergent container **150** based on the electrical signal received from the magnetic field sensor **300**. The controller **500** may control the user interface **15** to output information about the remaining amount of detergent in the detergent container based on the electrical signal received from the magnetic field sensor **300**.

[0168] The detergent pump **400** may move the liquid detergent accommodated in each of the first chamber **151** and the second chamber **152** to the mixing chamber **141**. The controller **500** may selectively move the detergents accommodated in each of the first chamber **151** and the second chamber **152** to the mixing chamber **141** based on the washing course and/or the user input. In addition, the controller **500** may sequentially move the detergents accommodated in each of the first chamber **151** and the second chamber **152** to the mixing chamber **141**.

[0169] The detergent detection electrode **430** may detect detergent in the detergent container **150**. A plurality of detergent detection electrodes **430** may be provided in each of the first chamber **151** and the second chamber **152** of the detergent container **150**. The controller **500** may apply current to the plurality of detergent detection electrodes **430** and detect a voltage between the plurality of detergent detection electrodes **430**. In a case where the detergent detection electrode **430** is in contact with detergent, the controller **500** may determine that detergent is present in the detergent container **150** based on the voltage being detected as a result of current applied to the plurality of detergent detection electrodes **430**.

[0170] The controller **500** may determine an electrical conductivity of detergent by applying a current having a defined frequency to the plurality of detergent detection electrodes **430**. The controller **500** may provide information about a state of the detergent through the user interface **15** based on the electrical conductivity of the detergent. The controller **500** may adjust the amount of detergent to be supplied to the tub **20** based on the electrical conductivity of the detergent. In addition, the controller **500** may determine the type of detergent based on the electrical conductivity of the detergent.

[0171] For example, the controller **500** may determine a first electrical conductivity of the detergent by applying a first current having a first frequency to the plurality of detergent detection electrodes **430**. The controller **500** may detect a first voltage between the plurality of detergent detection electrodes **430** according to the application of the first current. The controller **500** may determine the first electrical conductivity based on the first current and the first voltage.

[0172] The controller **500** may determine a second electrical conductivity of the detergent by applying a second current having a second frequency different from the first frequency to the plurality of detergent detection electrodes **430**. The controller **500** may detect a second voltage between the plurality of detergent detection electrodes **430** according to the application of the second current. The controller **500** may determine the second electrical conductivity based on the second current and the second voltage.

[0173] The first electrical conductivity of the detergent determined by applying the first current having the first frequency to the detergent detection electrodes **430** may vary depending on a concentration of the detergent. The controller **500** may determine the concentration of the detergent based on the first electrical conductivity of the detergent. The controller **500** may determine the detergent as laundry detergent or fabric softener based on the first electrical conductivity of the detergent.

[0174] The concentration of liquid may represent the density, viscosity, or thinness of the liquid. In general, the concentration of laundry detergent and the concentration of fabric softener are significantly different, and the concentration of fabric softener is relatively low. In other words, the viscosity of fabric softener is relatively low.

[0175] In addition, the second electrical conductivity of the detergent determined by applying the second current having the second frequency to the detergent detection electrodes **430** may vary depending on an acidity of the detergent. The controller **500** may determine the acidity of the detergent based on the second electrical conductivity of the detergent. Laundry detergents may be classified into alkaline detergents, neutral detergents, and acidic detergents. In general, alkaline detergents or neutral detergents are used for washing clothes, depending on the type of fabric. The controller **500** may determine the detergent as an alkaline detergent, a neutral detergent, or an acidic detergent based on the second electrical conductivity of the detergent.

[0176] The first frequency and the second frequency may be determined variously depending on the design. The first frequency and the second frequency may be different from each other. The first frequency may be higher than the second frequency. For example, the first frequency may be 300 KHz, and the second frequency may be 10 Hz. In other words, the first frequency of the first current applied to the detergent detection electrodes **430** to determine the concentration of the detergent is different from the second frequency of the second current applied to the detergent detection electrodes **430** to determine the acidity of the detergent.

[0177] As described above, the detergent container **150** may include the first chamber **151** and the second chamber **152**. The controller **500** may control the user interface **15** to provide information about whether the detergent accommodated in each of the first chamber and the second chamber is laundry detergent or fabric softener.

[0178] The controller **500** may control the user interface **15** to provide a chamber selection notification to guide selection of one of the first chamber **151** and the second chamber **152**, in a case where laundry detergent is accommodated in both the first chamber **151** and the second chamber **152**. The controller **500** may control the user interface **15** to provide a detergent error notification, in a case where fabric softener is accommodated in both the first chamber **151** and the second chamber **152**.

[0179] The controller **500** may control the user interface **15** to provide a detergent replenishment notification to notify detergent requires to be replenished based on a difference between the electrical conductivity (first electrical conductivity) and a reference electrical conductivity being greater than or equal to a defined reference value.

[0180] The controller **500** may adjust the amount of detergent to be supplied to the tub **20** according to the difference between the electrical conductivity (first electrical conductivity) and the reference electrical conductivity. The controller **500** may adjust the amount of detergent to be supplied to the tub **20** by adjusting a detergent supply time for supplying the detergent.

[0181] For example, in a case where the electrical conductivity is less than the reference electrical conductivity, the controller **500** may increase the detergent supply time to increase the amount of detergent to be supplied to the tub **20**. Conversely, in a case where the electrical conductivity is greater than the reference electrical conductivity, the controller **500** may decrease the detergent supply time to decrease the amount of detergent to be supplied to the tub **20**.

[0182] In a case where a washing process is required to be performed without replenishing the detergent despite the detergent replenishment notification provided, the controller **500** may adjust the amount of detergent to be supplied to the tub **20**. The controller **500** may also adjust the amount of detergent to be supplied to the tub **20**, based on the detergent not being replenished for a predetermined wait time after the detergent replenishment notification is provided.

[0183] A user may set whether to be provided with the detergent replenishment notification by operating the user interface **15** or the user device. In a case where the detergent replenishment notification is set to be disabled, the controller **500** may adjust the amount of detergent to be supplied to the tub **20** based on the electrical conductivity of the detergent, regardless of the provision of the detergent replenishment notification.

[0184] In addition, the controller **500** may control the user interface **15** to provide a detergent hardening notification to warn of detergent hardening, based on the electrical conductivity of the detergent being increased by a defined limit value over the reference electrical conductivity.

[0185] The controller **500** may reset the reference electrical conductivity based on the opening and closing of the detergent container **150**. The controller **500** may determine an electrical conductivity of the detergent, which is determined for the first time after the detergent container **150** is opened and then closed, as the reference electrical conductivity. The controller **500** may identify the opening and closing of the detergent container **150** based on an electrical signal transmitted from the detergent container open/close sensor **90**.

[0186] Information about a state of the detergent may include at least one of information about a concentration of the detergent, information about an acidity of the detergent, detergent replenishment notification, detergent error notification, detergent hardening notification, or chamber selection notification. The information about a state of the detergent may be provided through the user device as well as the user interface **15**. The controller **500** may control the communication interface **80** to transmit the information about a state of the detergent to the user device.

[0187] FIG. **8** is a flowchart briefly illustrating a method for controlling a washing machine according to an embodiment.

[0188] Referring to FIG. **8**, the controller **500** of the washing machine **1** may apply a current having a defined frequency to the plurality of detergent detection electrodes **430** (**801**). In FIG. **8**, the defined frequency may correspond to the first frequency described above, and the current may correspond to the first current described above. In a case where detergent is present in the detergent container **150**, the controller **500** may detect a voltage between the plurality of detergent detection electrodes **430**. The controller **500** may determine that detergent is present in the detergent container **150** based on detecting the voltage between the plurality of detergent detection electrodes **430**.

[0189] The controller **500** may determine an electrical conductivity of the detergent (**802**). The controller **500** may determine the electrical conductivity of the detergent based on the current applied to the plurality of detergent detection electrodes **430** and the voltage between the plurality of detergent detection electrodes **430** (**802**). In FIG. **8**, the electrical conductivity may correspond to the first electrical conductivity described above. The controller **500** may determine a concentration of the detergent corresponding to the electrical conductivity of the detergent.

[0190] The controller **500** may perform at least one of providing information about a state of the detergent through the user interface **15** based on the electrical conductivity of the detergent or adjusting the amount of the detergent to be supplied to the tub **20** (**803**). The information about the state of the detergent and the adjusting the amount of the detergent are further described in FIG. **9** and FIG. **10**.

[0191] FIG. **9** is a flowchart illustrating the method described in FIG. **8** in greater detail.

[0192] Operations **901** and **902** in FIG. **9** correspond to operations **801** and **802** in FIG. **8**.

[0193] The controller **500** of the washing machine **1** may compare an electrical conductivity of detergent with a reference electrical conductivity (**903**). The controller **500** may obtain a difference between the electrical conductivity of the detergent and the reference electrical conductivity. The controller **500** may control the user interface **15** to provide a detergent replenishment notification based on the difference between the electrical conductivity of the detergent and the reference electrical conductivity being greater than or equal to a defined reference value (**904**).

[0194] In a case where the difference between the electrical conductivity of the detergent and the reference electrical conductivity is less than the reference value, the controller **500** may determine that a concentration of the detergent is normal and that the detergent replenishment notification is not required. In this case, the controller **500** may control the user interface **15** to output information indicating that the concentration of the detergent is normal.

[0195] The difference between the electrical conductivity and the reference electrical conductivity may indicate the amount of increase in the electrical conductivity or the amount of decrease in the electrical conductivity. An increase in the electrical conductivity of the detergent may indicate that moisture in the detergent has evaporated and the concentration of the detergent has increased. Conversely, a decrease in the electrical conductivity of the detergent may indicate that moisture has been absorbed into the detergent and the concentration of the detergent has decreased. In both cases where the detergent concentration has increased and where the detergent concentration has decreased, detergent replenishment is required.

[0196] The evaporation of moisture from the detergent or the absorption of moisture into the detergent may occur for various causes. For example, in a case where the first chamber **151** and the second chamber **152** are not sealed by the detergent container cover **130**, the evaporation of moisture from the detergent or the absorption of moisture into the detergent may occur. Moisture may also be absorbed into the detergent due to the residual moisture after washing the detergent container **150**.

[0197] In a case where the concentration of the detergent increases, the detergent requires to be replenished to reduce the concentration of the detergent. In a case where the amount of relatively high concentration detergent is the same as the amount of detergent with a normal range concentration, the high concentration detergent contains more washing ingredients. Using the high concentration detergent in a washing process may cause an excessive amount of washing ingredients to enter the tub **20**, which may remain on the laundry after the washing process is completed.

[0198] In a case where the concentration of the detergent decreases, the detergent requires to be replenished to increase the concentration of the detergent. In a case where the amount of relatively low concentration detergent is the same as the amount of detergent with a normal range concentration, the low concentration detergent contains less washing ingredients. Accordingly, using the low concentration detergent in the washing process may result in an incomplete washing. In other words, an washing effect by the detergent may be reduced.

[0199] In order to maintain the concentration of the detergent appropriately, the washing machine **1** may notify a user that detergent requires to be replenished.

[0200] The controller **500** of the washing machine **1** may identify whether detergent in the detergent container **150** has been replenished (**905**). Whether detergent has been replenished may be identified by various methods. For example, the controller **500** may determine that detergent has been replenished based on identifying the opening and closing of the detergent container **150**. As another example, the controller **500** may determine that detergent has been replenished based on a decrease in an intensity of the magnetic field detected by the magnetic field sensor **300**. As described above, when detergent is replenished, the floating device **200** rises and the intensity of the magnetic field detected by the magnetic field sensor **300** decreases.

[0201] In a case where the detergent is replenished, the controller **500** may control the user interface **15** to stop displaying the detergent replenishment notification. For example, the detergent replenishment notification displayed through the user interface **15** may no longer be displayed as the detergent is replenished.

[0202] In a case where the detergent is replenished, the controller **500** may determine an electrical conductivity of the detergent by applying current to the plurality of detergent detection electrodes **430** again. That is, the controller **500** may periodically monitor the electrical conductivity of the detergent. The controller **500** may determine the electrical conductivity of the detergent at predetermined intervals.

[0203] In a case where the detergent is not replenished, the controller **500** may adjust a detergent supply time according to a difference between the electrical conductivity of the detergent and the reference electrical conductivity (**906**). That is, the controller **500** may adjust the amount of detergent to be supplied to the tub **20**.

[0204] The controller **500** may adjust the amount of detergent to be supplied to the tub **20**, in a case where a washing process is required to be performed without replenishing the detergent. The controller **500** may also adjust the amount of detergent to be supplied to the tub **20**, based on the detergent not being replenished for a predetermined wait time after the detergent replenishment notification is provided.

[0205] A user may set whether to be provided with the detergent replenishment notification by operating the user interface **15** or the user device. In a case where the detergent replenishment notification is set to be disabled, the controller **500** may adjust the amount of detergent to be supplied to the tub **20** based on the electrical conductivity of the detergent, regardless of the provision of the detergent replenishment notification.

[0206] FIG. **10** is a flowchart illustrating an embodiment of adjusting a detergent supply time described in FIG. **9** in greater detail.

[0207] Referring to FIG. **10**, in a case where the detergent is not replenished (No in operation **905**) even though the difference between the electrical conductivity of the detergent and the reference electrical conductivity is greater than or equal to the defined reference value, the controller **500** may adjust the amount of detergent to be supplied to the tub **20**. In FIG. **10**, non-replenishment of the detergent is exemplified as a precondition for adjusting the amount of the detergent, but the disclosure is not limited thereto. That is, because the detergent replenishment notification may be set to be disabled, the controller **500** may also adjust the amount of detergent to be supplied to the tub **20** in a case where the difference between the electrical conductivity of the detergent and the reference electrical conductivity is greater than or equal to the defined reference value (Yes in operation **903**).

[0208] In a case where the difference between the electrical conductivity of the detergent and the reference electrical conductivity is greater than or equal to the defined reference value and the electrical conductivity of the detergent is less than the reference electrical conductivity (No in operation **1001**), the controller **500** may increase the detergent supply time (**1002**). As the detergent supply time increases, the amount of detergent to be supplied to the tub **20** may increase. In a case where the concentration of the detergent is relatively low, the washing machine **1** may prevent a decrease in the washing effect by increasing the amount of detergent to be supplied to the tub **20**.

[0209] In a case where the difference between the electrical conductivity of the detergent and the reference electrical conductivity is greater than or equal to the defined reference value and the electrical conductivity of the detergent is greater than the reference electrical conductivity (Yes in operation **1001**), the controller **500** may reduce the detergent supply time (**1003**). As the detergent supply time is reduced, the amount of detergent to be supplied to the tub **20** may be reduced. In a case where the concentration of the detergent is relatively high, the washing machine **1** may prevent excessive washing ingredients from entering the tub **20** by reducing the amount of detergent to be supplied to the tub **20**.

[0210] As long as there is a difference between the electrical conductivity of the detergent and the reference electrical conductivity, the electrical conductivity of the detergent cannot be the same as the reference electrical conductivity. Accordingly, in FIG. **10**, a case where the electrical conductivity of the detergent and the reference electrical conductivity are the same is not considered.

[0211] In a case where moisture evaporation from the detergent continues, the electrical conductivity of the detergent may continue to increase. In a case where moisture evaporation from the detergent continues, the detergent may harden. The hardened detergent may not be put into the tub **20** when the detergent requires to be put in.

[0212] The controller **500** may identify whether the difference between the electrical conductivity of the detergent and the reference electrical conductivity is greater than or equal to a limit value (**1004**). The controller **500** may control the user interface **15** to provide a detergent hardening notification to warn of the detergent hardening based on the electrical conductivity of the detergent being increased by the defined limit value over the reference electrical conductivity (**1005**). In addition, the controller **500** may stop the operation of the washing machine **1** at the same time as the detergent hardening notification. By stopping the operation of the washing machine **1**, the washing process may be prevented from being performed without detergent.

[0213] The limit value used to determine whether the detergent is hardened may be greater than the reference value used to determine whether the detergent requires to be replenished.

[0214] FIG. **11** is a flowchart illustrating a method of determining the reference electrical conductivity and determining the type of detergent described in FIG. **9**.

[0215] Referring to FIG. **11**, the controller **500** may detect whether the detergent container **150** is opened and closed (**1101**). In a case where the opening and closing of the detergent container **150** is not detected, the controller **500** may determine that detergent is not added to the detergent container **150**. Accordingly, the controller **500** may apply a current having a defined frequency to the detergent detection electrodes **430** to determine an electrical conductivity of the detergent remaining in the detergent container **150** (**901**).

[0216] In response to detecting the opening and closing of the detergent container **150**, the controller **500** may reset the reference electrical conductivity (**1102**). In a case where the opening and closing of the detergent container **150** is detected, the controller **500** may determine that detergent has been added to the detergent container **150**. Because detergent is newly added to the detergent container **150**, the concentration of the detergent in the detergent container **150** may change. Accordingly, the reference electrical conductivity that serves as a criterion for determining the change in detergent concentration is required to be newly determined.

[0217] The controller **500** may determine a first electrical conductivity of the detergent by applying a first current having a first frequency to the plurality of detergent detection electrodes **430** (**1103**). The controller **500** may detect a first voltage between the plurality of detergent detection electrodes **430** according to the application of the first current. The controller **500** may determine the first electrical conductivity based on the first current and the first voltage.

[0218] In a case where the first electrical conductivity of the detergent is less than a defined threshold value (No in operation **1104**), the controller **500** may determine that the detergent in the detergent container **150** is fabric softener (**1105**). Conversely, the controller **500** may determine that the detergent in the detergent container **150** is laundry detergent (**1106**), in a case where the first electrical conductivity of the detergent is greater than or equal to the defined threshold value (Yes in operation **1104**).

[0219] The concentration of liquid may represent the density, viscosity, or thinness of the liquid. In general, the concentration of laundry detergent and the concentration of fabric softener are significantly different, and the concentration of fabric softener is relatively low. In other words, the viscosity of fabric softener is relatively low.

[0220] However, the detergent for infants may have a relatively low concentration despite being laundry detergent. The concentration of the detergent for infants may be similar to that of fabric softener. Accordingly, in the case of the detergent for infants, a user is required to separately input the type of detergent through the user interface **15** or the user device.

[0221] After determining the detergent as laundry detergent based on the first electrical conductivity of the detergent, the controller **500** of the washing machine **1** may determine a second electrical conductivity of the detergent (**1107**). The controller **500** may determine the second electrical conductivity of the detergent by applying a second current having a second frequency different from the first frequency to the plurality of detergent detection electrodes **430**. The controller **500** may determine an acidity of the detergent corresponding to the second electrical conductivity of the detergent (**1108**).

[0222] Laundry detergents may be classified into alkaline detergents, neutral detergents, and acidic detergents. In general, alkaline detergents or neutral detergents are used for washing clothes, depending on the type of fabric. The controller **500** may determine the detergent as an alkaline detergent, a neutral detergent, or an acidic detergent based on the second electrical conductivity of the detergent.

[0223] The controller **500** may determine the first electrical conductivity of the detergent, determined for the first time after the detergent container **150** is opened and closed, as the reference electrical conductivity (**1109**). That is, the first electrical conductivity of the detergent determined immediately after the detergent is added to the detergent container **150** may correspond to the reference electrical conductivity. Thereafter, the controller **500** may determine the change in detergent concentration by comparing the first electrical conductivity detected periodically with the reference electrical conductivity.

[0224] FIG. **12** is a graph showing electrical conductivities corresponding to various detergent concentrations.

[0225] As described above, the washing machine **1** may determine the electrical conductivity of the detergent in the detergent container **150**, and may distinguish whether the detergent is laundry detergent or fabric softener based on the determined electrical conductivity. The electrical conductivity shown in the graph **1200** of FIG. **12** corresponds to the first electrical conductivity determined by applying the first current having the first frequency to the plurality of detergent detection electrodes **430**.

[0226] Referring to the graph **1200** of FIG. **12**, different detergents may exhibit different electrical conductivities. For example, D**1**, D**2**, D**3**, D**4**, D**5**, D**6**, D**7**, and D**8** are laundry detergents produced by different manufacturers. S**1**, S**2**, and S**3** are fabric softeners produced by different manufacturers.

[0227] In general, the concentration of laundry detergent and the concentration of fabric softener are significantly different, and the concentration of fabric softener is relatively low. In other words, the viscosity of fabric softener is relatively low. As shown in the graph **1200** of FIG. **12**, the electrical conductivities of laundry detergents D**1**, D**2**, D**3**, D**4**, D**5**, D**6**, D**7**, and D**8** may all be greater than a defined threshold value EC_th. The electrical conductivities of fabric softeners S**1**, S**2**, and S**3** may all be less than the defined threshold value EC_th. The unit of electrical conductivity may be expressed as [uS/cm].

[0228] As such, the washing machine **1** may automatically identify the type of detergent by detecting the electrical conductivity, which is a unique property of the detergent in the detergent container **150**.

[0229] FIG. **13** is a graph showing first electrical conductivities corresponding to various detergent concentrations and second electrical conductivities corresponding to various detergent acidities.

[0230] Referring to a graph **1300** of FIG. **13**, the horizontal axis represents the first electrical conductivity described in the graph **1200** of FIG. **12**. The vertical axis represents the second electrical conductivity determined by applying the second current having the second frequency to the plurality of detergent detection electrodes **430**.

[0231] As described above, the washing machine **1** may determine the second electrical conductivity of the detergent, after determining that the detergent is laundry detergent based on the first electrical conductivity of the detergent. The washing machine **1** may distinguish the sub-types of the laundry detergent based on the second electrical conductivity of the detergent.

[0232] Laundry detergents may be classified into alkaline detergents, neutral detergents, and acidic detergents. Alkaline detergents or neutral detergents are generally used for washing clothes, depending on the type of fabric. The washing machine **1** may determine the detergent as an alkaline detergent, a neutral detergent, or an acidic detergent based on the second electrical conductivity of the detergent.

[0233] As shown in the graph **1300** of FIG. **13**, the second electrical conductivity of each detergent corresponding to D**1**, D**2**, D**4**, D**5**, D**6**, D**7**, and D**8** is all lower than EC**1**. D**1**, D**2**, D**4**, D**5**, D**6**, D**7**, and D**8** may all correspond to alkaline detergents. On the other hand, the second electrical conductivity of the detergent corresponding to D**3** is greater than EC**1** and lower than EC**2**. D**3** may correspond to a neutral detergent. “W” represents water. As such, the higher the acidity of the detergent, the higher the second electrical conductivity of the detergent may appear.

[0234] As described above, the washing machine **1** may distinguish the type of detergent as laundry detergent or fabric softener, and may distinguish the sub-types of the laundry detergent by applying currents having different frequencies to the detergent detection electrodes **430**.

[0235] FIG. **14** is a table illustrating example information provided for a detergent accommodated in a plurality of chambers of a detergent container.

[0236] Referring to a table **1400** of FIG. **14**, the washing machine **1** may provide various information about detergent through the user interface **15**. The information about detergent may be provided through a user device as well as through the user interface **15**.

[0237] As described above, the detergent container **150** may include the first chamber **151** and the second chamber **152**. The washing machine **1** may provide information about the detergent accommodated in each of the first chamber **151** and the second chamber **152**. The first chamber **151** and the second chamber **152** may accommodate laundry detergent or fabric softener.

[0238] In a case 1, laundry detergent is accommodated in the first chamber **151** and fabric softener is accommodated in the second chamber **152**. In this case, the washing machine **1** may identify that the detergent in the first chamber **151** is laundry detergent. In addition, the washing machine **1** may determine an acidity of the laundry detergent accommodated in the first chamber **151**. The washing machine **1** may identify that the detergent accommodated in the second chamber **152** is fabric softener. Accordingly, the user interface **15** may provide information about the detergent in the first chamber **151**, i.e., that the detergent is laundry detergent and an acidity of the laundry detergent. In addition, the user interface **15** may provide information about the detergent in the second chamber **152**, i.e., that the detergent is fabric softener. The washing machine **1** may control the detergent pump **400** to allow the laundry detergent in the first chamber **151** to be added to the tub **20** during a washing process, and may control the detergent pump **400** to allow the fabric softener in the second chamber **152** to be added to the tub **20** during a rinsing process.

[0239] In a case 2, laundry detergent is accommodated in both the first chamber **151** and the second chamber **152**. In this case, the washing machine **1** may identify that the detergent accommodated in both the first chamber **151** and the second chamber **152** is laundry detergent, and may determine an acidity of the laundry detergent accommodated in each chamber. Accordingly, the user interface **15** may provide information about the detergent in each of the first chamber **151** and the second chamber **152**, i.e., that the detergent is laundry detergent and the acidity of the laundry detergent. In addition, the user interface **15** may provide a chamber selection notification to guide a selection of one of the first chamber **151** and the second chamber **152**. That is, the washing machine **1** may guide a user to select laundry detergent to be used for the washing process. The washing machine **1** may control the detergent pump **400** to allow the laundry detergent accommodated in the chamber selected by the user to be added to the tub **20** during the washing process.

[0240] In a case 3, fabric softener is accommodated in the first chamber **151** and laundry detergent is accommodated in the second chamber **152**. In this case, the washing machine **1** may identify that the detergent in the first chamber **151** is fabric softener. The washing machine **1** may identify that the detergent in the second chamber **152** is laundry detergent. In addition, the washing machine **1** may determine an acidity of the laundry detergent accommodated in the second chamber **152**. Accordingly, the user interface **15** may provide information about the detergent in the first chamber **151**, i.e., that the detergent is fabric softener. In addition, the user interface **15** may provide information about the detergent in the second chamber **152**, i.e., that the detergent is laundry detergent and the acidity of the laundry detergent. The washing machine **1** may control the detergent pump **400** to allow the laundry detergent in the second chamber **152** to be added to the tub **20** during the washing process, and may control the detergent pump **400** to allow the fabric softener in the first chamber **151** to be added to the tub **20** during the rinsing process.

[0241] In a case 4, fabric softener is accommodated in both the first chamber **151** and the second chamber **152**. In this case, the washing machine **1** may identify that the detergent accommodated in both the first chamber **151** and the second chamber **152** is fabric softener. The user interface **15** may provide information about the detergent accommodated in each of the first chamber **151** and the second chamber **152**, i.e., that the detergent is fabric softener. In addition, the user interface **15** may provide a detergent error notification. The detergent error notification may allow the user to recognize that no laundry detergent is present in the detergent container **150**.

[0242] According to an embodiment of the disclosure, a washing machine may include: a tub; a detergent container configured to accommodate detergent to be supplied to the tub; a plurality of detergent detection electrodes configured to detect the detergent in the detergent container; a user interface configured to provide information about the washing machine; and a controller configured to be electrically connected to the plurality of detergent detection electrodes and the user interface. The controller may be configured to determine an electrical conductivity of the detergent by applying a current having a defined frequency to the plurality of detergent detection electrodes, and perform at least one of providing information about a state of the detergent through the user interface or adjusting an amount of detergent to be supplied to the tub based on the electrical conductivity.

[0243] The controller may be configured to control the user interface to provide a detergent replenishment notification indicating that the detergent requires to be replenished, based on a difference between the electrical conductivity and a reference electrical conductivity being greater than or equal to a defined reference value.

[0244] The controller may be configured to adjust the amount of detergent to be supplied to the tub according to a difference between the electrical conductivity and a reference electrical conductivity.

[0245] The controller may be configured to adjust the amount of detergent to be supplied to the tub by adjusting a detergent supply time for supplying the detergent.

[0246] The controller may be configured to increase the detergent supply time to increase the amount of detergent to be supplied to the tub, in response to the electrical conductivity being less than the reference electrical conductivity. The controller may be configured to decrease the detergent supply time to decrease the amount of detergent to be supplied to the tub, in response to the electrical conductivity being greater than the reference electrical conductivity.

[0247] The controller may be configured to control the user interface to provide a detergent hardening notification to warn of detergent hardening, based on the electrical conductivity being increased by a defined limit value over a reference electrical conductivity.

[0248] The controller may be configured to reset the reference electrical conductivity based on detecting opening and closing of the detergent container.

[0249] The controller may be configured to determine an electrical conductivity of the detergent, which is determined for the first time after the detergent container is opened and closed, as the reference electrical conductivity.

[0250] The defined frequency may be a first frequency, the current may be a first current, and the electrical conductivity may be a first electrical conductivity. The controller may be configured to determine a concentration of the detergent based on the first electrical conductivity of the detergent, and determine an acidity of the detergent based on a second electrical conductivity of the detergent determined by applying a second current having a second frequency different from the first frequency to the plurality of detergent detection electrodes.

[0251] The controller may be configured to determine the detergent as laundry detergent or fabric softener based on the concentration of the detergent.

[0252] The detergent container may include a first chamber and a second chamber. The controller may be configured to control the user interface to provide information about whether the detergent accommodated in each of the first chamber and the second chamber is the laundry detergent or the fabric softener.

[0253] The controller may be configured to control the user interface to provide a chamber selection notification to guide a selection of one of the first chamber and the second chamber, in a case where the laundry detergent is accommodated in both the first chamber and the second chamber.

[0254] The controller may be configured to control the user interface to provide a detergent error notification, in a case where the fabric softener is accommodated in both the first chamber and the second chamber.

[0255] According to an embodiment of the disclosure, in a method for controlling a washing machine including a tub, a detergent container configured to accommodate detergent, and a plurality of detergent detection electrodes configured to detect the detergent in the detergent container, the method may include: determining an electrical conductivity of the detergent by applying a current having a defined frequency to the plurality of detergent detection electrodes; and performing at least one of providing information about a state of the detergent through a user interface or adjusting an amount of detergent to be supplied to the tub based on the electrical conductivity.

[0256] The providing of the information about the state of the detergent may include providing a detergent replenishment notification indicating that the detergent requires to be replenished, based on a difference between the electrical conductivity and a reference electrical conductivity being greater than or equal to a defined reference value.

[0257] The amount of detergent may be adjusted according to a difference between the electrical conductivity and a reference electrical conductivity.

[0258] The adjusting of the amount of detergent may be performed by adjusting a detergent supply time for supplying the detergent.

[0259] The adjusting of the amount of detergent may include: increasing the detergent supply time to increase the amount of detergent to be supplied to the tub, in response to the electrical conductivity being less than the reference electrical conductivity; and decreasing the detergent supply time to decrease the amount of detergent to be supplied to the tub, in response to the electrical conductivity being greater than the reference electrical conductivity.

[0260] The providing of the information about the state of the detergent may include providing a detergent hardening notification to warn of detergent hardening, based on the electrical conductivity being increased by a defined limit value over a reference electrical conductivity.

[0261] The defined frequency may be a first frequency, the current may be a first current, and the electrical conductivity may be a first electrical conductivity. The method may further include determining a concentration of the detergent based on the first electrical conductivity of the detergent, and determining an acidity of the detergent based on a second electrical conductivity of the detergent determined by applying a second current having a second frequency different from the first frequency to the plurality of detergent detection electrodes.

[0262] According to the disclosure, the washing machine and the method for controlling the same may detect changes in concentration of detergent in a detergent container and provide a detergent replenishment notification based on the change in detergent concentration. In addition, the washing machine and the method for controlling the same may adjust the amount of detergent to be supplied to a tub according to changes in concentration of detergent in a detergent container. Through the above, the washing machine may minimize changes in washing performance due to the changes in detergent concentration.

[0263] According to the disclosure, the washing machine and the method for controlling the same may automatically identify the type of detergent in each of a plurality of chambers forming a detergent container and provide information about the detergent in each of the chambers. In addition, the washing machine and the method for controlling the same may automatically control operation of a detergent pump depending on the type of detergent accommodated in each of the chambers, and thus user convenience may be improved.

[0264] Meanwhile, the disclosed embodiments may be implemented in the form of a recording medium that stores instructions executable by a computer. The instructions may be stored in the form of program codes, and when executed by a processor, the instructions may create a program module to perform operations of the disclosed embodiments. The recording medium may be implemented as a computer-readable recording medium.

[0265] The computer-readable recording medium may include all kinds of recording media storing instructions that can be interpreted by a computer. For example, the computer-readable recording medium may be read only memory (ROM), random access memory (RAM), a magnetic tape, a magnetic disc, a flash memory, an optical data storage device, etc.

[0266] Also, the computer-readable recording medium may be provided in the form of a non-transitory storage medium, wherein the term 'non-transitory storage medium' simply means that the storage medium is a tangible device, and does not include a signal (e.g., an electromagnetic wave), but this term does not differentiate between where data is semi-permanently stored in the storage medium and where the data is temporarily stored in the storage medium. For example, a 'non-transitory storage medium' may include a buffer in which data is temporarily stored.

[0267] According to an embodiment of the disclosure, a method according to various embodiments of the disclosure may be included and provided in a computer program product. The computer program product may be traded as a product between a seller and a buyer. The computer program product may be distributed in the form of a machine-readable storage medium (e.g., compact disc read only memory (CD-ROM)), or be distributed (e.g., downloadable or uploadable) online via an application store (e.g., Play Store™) or between two user devices (e.g., smart phones) directly. When distributed online, at least part of the computer program product (e.g., a downloadable app) may be temporarily generated or at least temporarily stored in the machine-readable storage medium, such as a memory of the manufacturer's server, a server of the application store, or a relay server.

[0268] Although embodiments of the disclosure have been described with reference to the accompanying drawings, a person having ordinary skilled in the art will appreciate that other specific modifications may be easily made without departing from the technical spirit or essential features of the disclosure. Accordingly, the foregoing embodiments should be regarded as illustrative rather than limiting in all aspects.

1. A washing machine, comprising:
- a tub;
- a detergent container configured to accommodate detergent to be supplied to the tub;
- a plurality of electrodes in the detergent container;
- a user interface configured to provide information to a user; and
- a controller configured to, with detergent accommodated in the detergent container:
  - apply a current having a defined frequency to electrodes of the plurality of electrodes,
  - detect a voltage between electrodes of the plurality of electrodes,
  - determine an electrical conductivity of the detergent based on the detected voltage, and
  - provide, based on the determined electrical conductivity, information about a state of the detergent through the user interface and/or adjust an amount of the detergent to be supplied to the tub.

2. The washing machine of claim **1**, wherein the controller is further configured to, with the detergent accommodated in the detergent container, provide, based on a difference between the determined electrical conductivity and a reference electrical conductivity being greater than or equal to a defined reference value, a detergent replenishment notification through the user interface indicating that replenishment of the detergent is required.

3. The washing machine of claim **1**, wherein the controller is further configured to, with the detergent accommodated in the detergent container, adjust the amount of the detergent to be supplied to the tub based on a difference between the determined electrical conductivity and a reference electrical conductivity.

4. The washing machine of claim **3**, wherein the controller is further configured to, with the detergent accommodated in the detergent container, adjust the amount of the detergent to be supplied to the tub by adjusting a detergent supply time for supplying the detergent to the tub.

5. The washing machine of claim **4**, wherein
 the controller is further configured to, with the detergent accommodated in the detergent container:
 increase the detergent supply time to increase the amount of the detergent to be supplied to the tub, in response to the determined electrical conductivity being less than the reference electrical conductivity, and
 decrease the detergent supply time to decrease the amount of the detergent to be supplied to the tub, in response to the determined electrical conductivity being greater than the reference electrical conductivity.

6. The washing machine of claim **1**, wherein the controller is further configured to, with the detergent accommodated in the detergent container, provide, based on the determined electrical conductivity being increased by a defined limit value over a reference electrical conductivity, a detergent hardening notification through the user interface to warn that the detergent is hardened.

7. The washing machine of claim **2**, wherein
 the controller is further configured to:
 detect an opening and closing of the detergent container, and
 set the reference electrical conductivity based on the opening and closing of the detergent container being detected.

8. The washing machine of claim **2**, wherein
 the controller is further configured to:
 detect an opening and closing of the detergent container set the reference electrical conductivity to the electrical conductivity determined after the detergent container is opened and closed.

9. The washing machine of claim **1**, wherein
 the defined frequency is a first frequency, the current is a first current, the detected voltage is a first voltage, and the determined electrical conductivity is a first electrical conductivity, and
 the controller is further configured to, with the detergent accommodated in the detergent container:
 determine a concentration of the detergent based on the first electrical conductivity,
 apply a second current having a second frequency different from the first frequency to electrodes of the plurality of electrodes,
 detect a second voltage between electrodes of the plurality of electrodes,
 determine a second electrical conductivity of the detergent based on the detected second voltage, and
 determine an acidity of the detergent based on the determined second electrical conductivity.

10. The washing machine of claim **9**, wherein the controller is further configured to, with the detergent accommodated in the detergent container, determine whether the detergent is laundry detergent or fabric softener based on the determined concentration.

11. The washing machine of claim **10**, wherein
 the detergent container includes a first chamber and a second chamber, and
 the controller is further configured to, with the detergent accommodated in the first chamber and the second chamber, provide information through the user interface about whether the detergent accommodated in each of the first chamber and the second chamber is laundry detergent or fabric softener.

12. The washing machine of claim **11**, wherein the controller is further configured to provide a chamber selection notification through the user interface to guide a selection of one of the first chamber and the second chamber, in response to the laundry detergent being accommodated in both the first chamber and the second chamber.

13. The washing machine of claim **11**, wherein the controller is further configured to provide a detergent error notification through the user interface in response to the fabric softener being accommodated in both the first chamber and the second chamber.

14. A method for controlling a washing machine including a tub, a detergent container configured to accommodate detergent to be supplied to the tub, a plurality of electrodes in the detergent container, and a user interface configured to provide information to a user, the method comprising:
 with detergent accommodated in the detergent container,
 applying a current having a defined frequency to electrodes of the plurality of electrodes,
 detecting a voltage between electrodes of the plurality of electrodes,
 determining an electrical conductivity of the detergent based on the detected voltage, and
 providing, based on the determined electrical conductivity, information about a state of the detergent through the user interface and/or adjusting an amount of the detergent to be supplied to the tub.

15. The method of claim **14**, wherein the providing includes providing, based on a difference between the determined electrical conductivity and a reference electrical conductivity being greater than or equal to a defined reference value, a detergent replenishment notification indicating that replenishment of the detergent is required.

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